# Supplementary material for

A Lightweight Detection Pipeline for Earthquake Early Warning and Its Evaluation on Three Seismic Networks

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Every number below was produced by a tool in the pipeline's repository (`https://github.com/hogib/ayzek`, `tools/`) or in the offline evaluation code, and the command is named with each table. The AFAD waveforms cannot be redistributed; the KOERI datasets are attached to the pipeline's releases. Sections S1 to S19 concern the development data, on which every choice of the method was made (main-text Section 3.1); the AFAD Sındırgı $M_w$ 6.1 and Simav sequences, called "held out" in the development record, are development data under the test protocol. Sections S20 to S23 hold the development results in full, the selected test sets and the test results that the main text summarizes.

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## S1 Stations and quiet windows

Table S1a lists the stations of the replays. All TU stations record HH channels at 100 Hz. No TU station contributed a window to the training, validation or test sets of the pipeline's detector; the KOERI datasets do not share this guarantee, and the role of each KO station in the detector's station split is given. The replay of the Sındırgı 30-minute window uses DEMI, MANT and BAND only. The eleventh station of the continuous archive, GCAM, is in none of the replays and is not listed.

Table S1b lists the 23 AFAD quiet windows, in which no cataloged event lies within 250 km of any of the eight stations during the window or for 30 min on either side of it. Each window is replayed on its own; its first 30 min warm the noise baseline and the STA/LTA and are not scored.

Reproduction: `tools/extract_quiet.py` selects the windows, and stage 6 of `tools/reproduce_all.sh` replays them with both detectors. Station coordinates: `models/stations.csv` and each KO dataset's `stations.json`.

**Table S1:** Stations and quiet windows.

**(a)** Stations. Datasets: S30, Sındırgı 30 min (tuning); Mar, Marmara (tuning); S61, Sındırgı $M_w$ 6.1; Sim, Simav $M_w$ 4.8 (both development sequences); Q, the AFAD quiet windows; and the three KOERI datasets.

| Station | Net. | Lat. | Lon. | Datasets | Detector split |
|---|---|---|---|---|---|
| ARNA | TU | 41.3470 | 28.6838 | Mar | none |
| BAND | TU | 40.3553 | 28.0992 | S30, Mar, Sim, Q | none |
| CATL | TU | 41.3407 | 28.3561 | Mar | none |
| CMH | TU | 40.0115 | 27.9703 | S61, Q | none |
| DEMI | TU | 39.0428 | 28.7162 | S30, Mar, S61, Sim | none |
| ELBA | TU | 41.1469 | 28.4307 | Mar, Q | none |
| KAND | TU | 41.0943 | 30.1922 | S61, Sim, Q | none |
| KURT | TU | 40.5905 | 29.2092 | Sim | none |
| KIRK | TU | 41.7817 | 27.0972 | Mar, S61, Q | none |
| MANT | TU | 38.4908 | 28.5579 | S30, Mar, Sim, Q | none |
| SEMS | TU | 40.8710 | 29.7391 | Mar, S61, Q | none |
| VIZE | TU | 41.5785 | 27.8400 | S61, Sim, Q | none |
| ARMT | KO | 40.5683 | 28.8660 | marmara_ko | test |
| BGKT | KO | 41.1810 | 28.7730 | marmara_ko | training |
| BOTS | KO | 40.9902 | 27.9810 | marmara_ko | training |
| CTKS | KO | 41.2373 | 28.5072 | marmara_ko | training |
| DKL | KO | 39.0713 | 26.9053 | demo_ko, quiet_ko | training |
| EDC | KO | 40.3465 | 27.8618 | marmara_ko | validation |
| GEDZ | KO | 39.0445 | 29.4105 | quiet_ko | test |
| KCTX | KO | 40.2627 | 28.3353 | demo_ko | training |
| KULA | KO | 38.5145 | 28.6607 | demo_ko, quiet_ko | test |
| MRMT | KO | 40.6090 | 27.5832 | marmara_ko | training |
| SIMA | KO | 39.0833 | 28.9825 | demo_ko, quiet_ko | validation |
| SLVT | KO | 41.2300 | 28.2100 | marmara_ko | training |
| TKR | KO | 40.9902 | 27.5357 | marmara_ko | test |
| TVSB | KO | 39.4497 | 29.4615 | demo_ko, quiet_ko | training |
| YAYO | KO | 39.6113 | 27.0726 | demo_ko, quiet_ko | training |

**(b)** AFAD quiet windows: scored interval (UTC; $^{+1}$, the next day), scored hours, stations (St.), and the alarms of the learned detector (L) and of the STA/LTA at equal unmatched alarms (S).

| Date | From | To | Hours | St. | L | S |
|---|---|---|---|---|---|---|
| 2024-05-22 | 19:54 | 02:05$^{+1}$ | 6.19 | 8 | 1 | 46 |
| 2024-05-23 | 04:22 | 07:48 | 3.43 | 8 | 1 | 11 |
| 2024-05-23 | 21:51 | 01:17$^{+1}$ | 3.43 | 8 | 2 | 28 |
| 2024-05-24 | 15:20 | 19:24 | 4.07 | 8 | 0 | 16 |
| 2024-05-26 | 00:31 | 03:48 | 3.27 | 8 | 0 | 17 |
| 2024-05-29 | 07:08 | 10:12 | 3.06 | 7 | 1 | 16 |
| 2024-05-30 | 15:45 | 20:57 | 5.21 | 7 | 1 | 14 |
| 2024-06-02 | 23:30 | 07:21$^{+1}$ | 7.85 | 8 | 2 | 30 |
| 2024-06-05 | 02:10 | 11:46 | 9.60 | 8 | 2 | 52 |
| 2024-06-06 | 10:54 | 13:56 | 3.04 | 8 | 0 | 11 |
| 2024-06-06 | 15:26 | 20:41 | 5.24 | 8 | 0 | 14 |
| 2024-06-09 | 12:40 | 18:05 | 5.41 | 8 | 2 | 11 |
| 2024-12-25 | 16:47 | 21:06 | 4.31 | 8 | 0 | 2 |
| 2025-01-04 | 20:37 | 23:29 | 2.87 | 8 | 0 | 1 |
| 2025-02-24 | 12:13 | 15:23 | 3.15 | 7 | 0 | 4 |
| 2025-02-26 | 16:55 | 20:25 | 3.50 | 8 | 2 | 10 |
| 2025-02-26 | 21:55 | 02:58$^{+1}$ | 5.06 | 8 | 0 | 1 |
| 2025-02-27 | 12:27 | 16:26 | 3.97 | 8 | 1 | 6 |
| 2025-02-27 | 18:41 | 00:14$^{+1}$ | 5.55 | 8 | 0 | 5 |
| 2025-03-02 | 14:54 | 17:51 | 2.94 | 7 | 1 | 8 |
| 2025-03-03 | 18:29 | 22:58 | 4.49 | 7 | 0 | 0 |
| 2025-03-04 | 19:00 | 22:12 | 3.19 | 7 | 0 | 6 |
| 2025-03-07 | 20:22 | 00:20$^{+1}$ | 3.96 | 7 | 0 | 14 |
| total | | | 102.79 | | 16 | 323 |

## S2 Verification and the data path

**Numerical agreement.** Every network of the pipeline is a transcription of its PyTorch module and reads the same `state_dict`. The test suite (15 tests, `meson test`) compares each stage with its reference on real waveforms (Table S2a). Window conditioning runs in double precision as SciPy does; the residual of the cleaned signal is the rounding of an eighth-order IIR filter in transfer-function form. The picker's picks are identical to PyTorch's. The aarch64 build passes the same suite under emulation and, on a Raspberry Pi 4, raises the same alarms at the same times with the same locations as the x86 build on the KOERI datasets.

**Undecodable records.** During the $M_w$ 6.2 of 2025-04-23, the dataloggers at CATL (57 km) and ARNA (68 km) switched from Steim-2 compression to uncompressed 32-bit integer records (CATL little-endian, ARNA big-endian); ELBA, at 38 km, did not. A decoder that accepted only Steim-2 rejected all of them, 104 records at CATL and 155 at ARNA, at 09:49–09:51 and 10:02–10:03 UTC, and the two stations therefore recorded no detection of the strongest shaking. The pipeline now decodes integer records in either byte order (both files decode bit-exactly against ObsPy) and reports, with the reason, any record it cannot decode. A scan of the whole archive (its 13 stations, the eleven of the continuous archive and ARNA and CATL; 283 chunks, 398 million records, 2024–2026) found 304 non-Steim-2 records, all at ARNA and CATL on that day between 09:49 and 12:13 UTC.

**Sample loss under load.** The pipeline dates a sample by its position in a ring of $2^{17}$ samples (21.8 min at 100 Hz) per component. An earlier implementation dropped the samples it could not hold when a ring was full. A network-wide archive outage of 2,095.6–2,103.4 s per component (1.60 times the ring, on all 27 components of nine stations at once) exposed the defect. Because position is ring index plus base, a dropped sample does not leave a hole but moves every later sample of that component earlier, and after the outage the three components of a station no longer describe the same instant. In the current pipeline, writes never wait and never drop. What does not fit is queued per component, and a gap is queued as a count, so an outage of any length costs no memory. Table S2b repeats the Marmara replay with a 25-minute hole cut into the record of all eight stations, ending 49 min before the $M_w$ 6.2. With the current pipeline the hole changes nothing but the two single-station detections inside it; with the earlier one, 329,439 dropped samples cost 15 of the 44 detections and the $M_w$ 5.9. The 11.6-hour window containing the outage is replayed without a stall or a drop, and yields exactly the windows, triggers, picks and alarms of the two runs split around it (711,760 windows, 86 triggers and the same five alarms to the hundredth of a second); only the windows rejected for crossing the gap differ.
Reproduction: `tests/` (`test_dsp`, `test_models`, `test_stalta`, `test_steim`, `test_writer`, …); `tools/scan_encodings.py` for the archive; the gapped replays are described in the repository's `docs/impl/05-pipeline.md`.

**Table S2:** Verification.

**(a)** Largest difference from the reference, on real windows.

| Stage | Reference | Largest difference |
|---|---|---|
| cleaned window (detrend, taper, 1–45 Hz zero-phase) | SciPy | $1.1 \times 10^{-8}$ relative |
| standardized window (float) | NumPy | $1.9 \times 10^{-9}$ absolute |
| detector: convolutional stack | PyTorch | $3.8 \times 10^{-6}$ |
| detector: BiLSTM | PyTorch | $1.3 \times 10^{-6}$ |
| detector: attention | PyTorch | $9.5 \times 10^{-7}$ |
| detector: logit | PyTorch | $6.0 \times 10^{-7}$ |
| detector: ensemble probability | PyTorch | identical to 5 decimals |
| picker: logits | PyTorch | $5.7 \times 10^{-6}$ (picks identical) |
| STA/LTA: filter coefficients | SciPy `butter(4, sos)` | $2 \times 10^{-16}$ |
| STA/LTA: filtered trace (peak $9 \times 10^5$ counts) | SciPy | $10^{-8}$ counts |
| STA/LTA: ratio, vertical | ObsPy `recursive_sta_lta` | $5 \times 10^{-14}$ relative |
| STA/LTA: ratio, three components | ObsPy | $8 \times 10^{-14}$ relative |

**(b)** Marmara replay (57 cataloged events), records starting 08:35–09:00 UTC removed from all eight stations.

| | Unbroken, current | Gap, current | Gap, earlier |
|---|---|---|---|
| stalls / samples dropped | 0 / 0 | 0 / 0 | 112 / 329,439 |
| alarms raised | 68 | 68 | 76 |
| alarms matching a cataloged event | 44 | 44 | 29 |
| alarms without one | 24 | 24 | 47 |
| cataloged events detected | 44 | 44 | 29 |
| $M_w$ 6.2 alarm | +11.0 s, ARNA+ELBA | identical | two alarms, CATL+ELBA |
| $M_w$ 5.9 alarm | +12.5 s | identical | not raised |

## S3 Quiet-window detections per station

Station detections on the AFAD quiet windows, after the lead-in of each window, with detections within 40 s at one station counted once. The rate is per day of the scored hours of the windows the station is part of; alarms are those the station declared. KIRK's detections under the learned detector are the instrumental pulse described in the main text (218 of the 225 at one phase of a 10-minute cycle locked to UTC). Under STA/LTA, CMH and KIRK dominate, and KAND, with 16 detections under the learned detector, does not trigger at all.

**Table S3:** Detections and alarms per station on 102.8 h of AFAD quiet data.

| | | Learned detector | | | STA/LTA, equal unmatched alarms | | |
|---|---|---|---|---|---|---|---|
| Station | Hours | Detections | Per day | Alarms | Detections | Per day | Alarms |
| BAND | 102.8 | 31 | 7.2 | 9 | 172 | 40.2 | 50 |
| CMH | 102.8 | 30 | 7.0 | 6 | 993 | 231.9 | 227 |
| ELBA | 102.8 | 1 | 0.2 | 0 | 30 | 7.0 | 4 |
| KAND | 88.2 | 16 | 4.4 | 2 | 0 | 0.0 | 0 |
| KIRK | 102.8 | 225 | 52.5 | 10 | 804 | 187.7 | 232 |
| MANT | 94.5 | 15 | 3.8 | 1 | 83 | 21.1 | 39 |
| SEMS | 99.6 | 33 | 7.9 | 3 | 153 | 36.9 | 56 |
| VIZE | 102.8 | 3 | 0.7 | 1 | 146 | 34.1 | 38 |
| total | | 354 | | 16 alarms | 2,381 | | 323 alarms |

Reproduction: `python3 tools/quiet_stations.py --runs learned=LEARNED_RUNS stalta=STALTA_RUNS`, with the two detectors' quiet-window run directories. An alarm is declared by two stations, so the per-station alarm counts sum to twice the alarms.

## S4 Computational cost

Each stage was timed with the pipeline's benchmark (`ayzek_bench`) on synthetic seeded input, after a warm-up, for at least one second and ten calls; inference time does not depend on the sample values. The x86 machine is an AMD Ryzen 5 5600 (6 cores, 12 threads), release build with GCC 16.2, idle; the Raspberry Pi 4 (Cortex-A72, 4 cores at 1.8 GHz) runs the static aarch64 build with NEON kernels, each group starting below 55 °C with the clock held at 1.8 GHz throughout. A station needs one window every 0.5 s; the capacity is the stations whose windows one to all cores sustain. The published models of main-text Section 3.2 were timed separately under PyTorch, one forward pass per 0.5 s step and station as the evaluation ran them.

**Table S4:** Cost per stage (median) and real-time capacity.

**(a)** Per call.

| Stage | Ryzen 5 5600 | Raspberry Pi 4 |
|---|---:|---:|
| condition a 6 s window | 0.04 ms | 0.14 ms |
| detector, one network | 1.7 ms | 11.8 ms |
| detector ensemble, three networks | 5.2 ms | 36.3 ms |
| STA/LTA, 50 samples (one 0.5 s step) | 0.0005 ms | 0.0019 ms |
| per 0.5 s step and station (conditioning, ensemble, STA/LTA) | 5.2 ms | |
| picker, one 60 s window | 12.0 ms | 75.7 ms |
| EQTransformer, one pass (60 s input), PyTorch, one thread | 9.1 ms | |
| PhaseNet, one pass (30 s input), PyTorch, one thread | 1.6 ms | |

**(b)** Stations whose detector windows the cores sustain.

| Threads | Ryzen 5 5600 | Raspberry Pi 4 |
|---|---:|---:|
| 1 | 95 | 14 |
| 4 | 379 | 52 |
| 12 | 653 | |

Reproduction: `build-release/bench/ayzek_bench --group stages,capacity`; `tools/bench_pi.sh` on the board; results in the repository's `bench/results/models/`.

## S5 Station subsets and alert time

The Marmara replay was recorded once (`ayzek --record`) and its network stage rerun for all 247 subsets of two to eight of its stations (`tools/network_subsets`), which gives the same result as replaying each subset. Every subset declares the $M_w$ 6.2, and the best subset of every size alerts at 11.0 s. Table S5a gives the best subset per size and Table S5b the mean change when a station is added to a subset without it. The four stations within 70 km each shorten the $M_w$ 6.2 alert by about 7 s; SEMS and CATL add the most declared events and also the most declared events without a catalog counterpart.

Against a geometric model in which the alert is the P travel time to the subset's second-nearest station plus a latency, the observed alerts of the 247 subsets give a latency of median 2.05 s (interquartile range 0.58–2.77 s) and a rank correlation of 0.71 between predicted and observed alert. The largest residuals, 5.7 s, are the subsets in which KIRK (140 km) is the second station, because KIRK stays below the 0.9 plateau for the $M_w$ 6.2 and declares through the persistence branch.

**Table S5:** Station subsets of the Marmara replay (57 cataloged events).

**(a)** Best subset per size, by the $M_w$ 6.2 alert.

| $k$ | Stations | $M_w$ 6.2 alert | Events declared | Without catalog event |
|---|---|---|---|---|
| 2 | ARNA, ELBA | 11.0 s | 19 | 1 |
| 3 | ARNA, ELBA, SEMS | 11.0 s | 37 | 4 |
| 4 | ARNA, ELBA, KIRK, SEMS | 11.0 s | 41 | 8 |
| 5 | ARNA, CATL, ELBA, KIRK, SEMS | 11.0 s | 44 | 21 |
| 6 | ARNA, BAND, DEMI, ELBA, KIRK, SEMS | 11.0 s | 44 | 14 |
| 7 | all but CATL | 11.0 s | 44 | 15 |
| 8 | all | 11.0 s | 44 | 24 |

**(b)** Mean change when the station is added, over the 120 subsets of two to seven stations without it.

| Station | Events declared | Without catalog event | $M_w$ 6.2 alert |
|---|---|---|---|
| ARNA | +3.40 | +0.59 | −7.4 s |
| BAND | +6.68 | +3.26 | −6.9 s |
| CATL | +7.54 | +8.45 | −7.1 s |
| DEMI | +1.52 | +1.51 | −0.4 s |
| ELBA | +7.19 | +2.18 | −7.4 s |
| KIRK | +4.62 | +2.99 | −0.9 s |
| MANT | +0.23 | +0.57 | 0.0 s |
| SEMS | +11.72 | +8.64 | −1.5 s |

Reproduction: `ayzek --scores-in CACHE --record marmara.tsv data/marmara/*.mseed`; `build-release/tools/network_subsets marmara.tsv --catalog tests/catalogs/marmara.csv`; `tools/station_geometry.py --stations ARNA,BAND,CATL,DEMI,ELBA,KIRK,MANT,SEMS --at 40.84373,28.21539,4.91 --compare subsets.csv`.

## S6 Association slack

Two detections may belong to one event if their P times differ by no more than the P travel time between the stations plus a slack, 3 s by default. The slack was sized for the P estimate before anchoring, good to about 2.5 s; anchored P times are good to about 0.1 s, so a tighter slack might be expected to reject chance pairings. It does not. No count changes by more than one event or one alarm (Table S6), because the admissible difference is dominated by the inter-station travel time, 7 s for ARNA–ELBA and about 50 s for the widest pairs of these networks.

**Table S6:** Association slack, two stations: detected / chance-corrected / unmatched alarms.

| Slack | Sındırgı 30 min (9 events) | Marmara (57) | Held out (66) |
|---|---|---|---|
| 3 s (default) | 7 / 7.0 / 2 | 44 / 41.7 / 24 | 39 / 36.4 / 7 |
| 1.5 s | 7 / 7.0 / 2 | 44 / 41.7 / 25 | 38 / 35.4 / 7 |
| 0.5 s | 7 / 7.0 / 2 | 45 / 43.0 / 25 | 38 / 35.4 / 7 |

Reproduction (both this table and Table S7): `tools/sweep_trigger.py --dataset NAME:CACHE:FILES --set 'slack=3 1.5 0.5' --set 'min-stations=2 3' --shifts 127,263,391`, with the window probabilities of stage 7 of `tools/reproduce_all.sh`.

## S7 Three-station declaration

Requiring a third station halves the unmatched alarms of the dense Marmara network for 1.2 s of median delay and 0.5 s on the $M_w$ 6.2. In the sparse Sındırgı $M_w$ 6.1 network the third station is 230–290 km away, and the $M_w$ 6.1 alarm comes 20 s later (Table S7). Two stations remain the default; three are worth requiring only where the network is dense. A larger reduction of unmatched alarms would need the declaration to test whether the detections fit one origin, rather than pairwise time differences.

**Table S7:** Two against three stations (slack 3 s).

| | Sındırgı 30 min | | Marmara | | Held out | |
|---|---|---|---|---|---|---|
| | 2 | 3 | 2 | 3 | 2 | 3 |
| detected | 7 | 5 | 44 | 38 | 39 | 25 |
| chance-corrected | 7.0 | 5.0 | 41.7 | 35.2 | 36.4 | 23.1 |
| unmatched alarms | 2 | 0 | 24 | 12 | 7 | 1 |
| median alarm delay | 18.0 s | 26.0 s | 12.0 s | 13.2 s | 17.5 s | 42.0 s |
| largest event | $M_w$ 4.9 +18.0 s | $M_w$ 4.9 +22.5 s | $M_w$ 6.2 +11.0 s | $M_w$ 6.2 +11.5 s | $M_w$ 6.1 +17.0 s | $M_w$ 6.1 +37.0 s |

## S8 Published models: adapter, selection and details

**Adapter.** `tools/seisbench_scores.py` (SeisBench 0.12.6) writes, for every 6 s window of the pipeline's 0.5 s grid, the largest model output within the window, which is the detection trace of EQTransformer or the P probability of PhaseNet. In the causal mode the model input ends at the window's end (60 s for EQTransformer, 30.01 s for PhaseNet); in the look-ahead mode it is centered on the window. Each model applies its own normalization; EQTransformer's original weights carry a 1–45 Hz band-pass, applied causally in the causal mode and zero-phase in the look-ahead mode. The scores replace the detector through `--scores-in`; everything downstream is the pipeline's. Every replay was checked to take every window from these scores (detector time 0.001–0.002 ms per window on every station).

**Selection.** The protocol, the candidate weights and the two rules were committed before the first tuning run (91b9ee1 of the paper's repository, 22 October 2026; the one adapter check that preceded it is recorded in 571ef5f). The candidates were the original, STEAD, INSTANCE and ETHZ weights of EQTransformer and the STEAD, INSTANCE and ETHZ weights of PhaseNet. The trigger grid on the two tuning replays crossed thresholds of 0.05, 0.1, 0.2, …, 0.9 with a persistence of 1, 2, 4 or 8 windows, with the instant branch off and the chance correction applied. Table S8a gives the selection. For the record, the best setting of the other weight sets at most 26 unmatched alarms reached 15.9 (EQTransformer ETHZ), 3.5 (original) and none (STEAD) chance-corrected detections in the causal mode, and PhaseNet ETHZ 45.4 and STEAD 31.4. Table S8b and c add, for the evaluation run once with these settings, the Sındırgı $M_w$ 6.1 magnitude bands and alarm delays and the KOERI datasets one by one; marmara_ko overlaps the detector's training data (134 windows at four stations) and demonstrates the pipeline without comparing detectors.

Look-ahead EQTransformer missed the $M_w$ 6.1 because its detection output at CMH peaked at 0.40 during the event, below its threshold of 0.5, and only DEMI declared. In the causal mode the same weights reached 0.97 at CMH. Reproduction: `tools/seisbench_scores.py`, `tools/select_setting.py`, and stage 8 of `tools/reproduce_all.sh` (`SEISBENCH=1` recomputes the scores).

With blocks of origin time instead of single events, the advantage of look-ahead PhaseNet's equal-unmatched setting becomes significant over the three AFAD replays together ($-10.8$, interval $-20.4$ to $-1.4$ with 10-minute blocks) and, with 15-minute blocks, on the Sındırgı $M_w$ 6.1 replay alone ($-11.9$ to $-0.2$); it remains a setting with three times the unmatched alarms that cannot run in real time.

**Table S8:** Published models in place of the pipeline's detector.

**(a)** Selected on the tuning replays (66 events; the learned detector: 51 detected, 48.7 chance-corrected, 26 unmatched alarms). All selected weights are INSTANCE.

| Model, mode | Rule | Threshold × windows | Detected / corrected | Unmatched |
|---|---|---|---|---|
| EQTransformer, causal | equal unmatched | $0.8 \times 2$ | 34 / 30.2 | 25 |
| EQTransformer, causal | equal detections | $0.3 \times 8$ | 58 / 55.3 | 72 |
| EQTransformer, look-ahead | both | $0.5 \times 8$ | 53 / 50.7 | 26 |
| PhaseNet, causal | both | $0.7 \times 2$ | 51 / 49.0 | 19 |
| PhaseNet, look-ahead | equal unmatched | $0.4 \times 8$ | 55 / 53.1 | 26 |
| PhaseNet, look-ahead | equal detections | $0.6 \times 8$ | 52 / 49.8 | 23 |

**(b)** Sındırgı $M_{\mathrm{w}}$ 6.1 replay: detected per magnitude band, median alarm delay and the $M_{\mathrm{w}}$ 6.1 alarm. Look-ahead delays are in parentheses: their input extends past the window and they cannot be achieved in real time.

| | M2–3 | M3–4 | M4–5 | Median delay | $M_{\mathrm{w}}$ 6.1 |
|---|---|---|---|---|---|
| learned | 12/27 | 21/32 | 5/6 | 17.5 s | +17.0 s |
| EQTransformer, causal, equal unmatched | 2/27 | 8/32 | 3/6 | 21.5 s | +17.0 s |
| EQTransformer, causal, equal detections | 15/27 | 21/32 | 4/6 | 30.5 s | +20.0 s |
| EQTransformer, look-ahead | 11/27 | 19/32 | 6/6 | (20.5 s) | missed |
| PhaseNet, causal | 7/27 | 22/32 | 6/6 | 17.8 s | +16.5 s |
| PhaseNet, look-ahead, equal unmatched | 17/27 | 23/32 | 5/6 | (20.0 s) | (+19.0 s) |
| PhaseNet, look-ahead, equal detections | 12/27 | 23/32 | 5/6 | (20.0 s) | (+19.5 s) |

**(c)** KOERI datasets: detected / unmatched alarms; quiet_ko (5.8 h): alarms; the $M_{\mathrm{w}}$ 6.2 alarm on marmara_ko.

| | demo_ko | marmara_ko | quiet_ko | $M_{\mathrm{w}}$ 6.2 |
|---|---|---|---|---|
| learned | 8/9, 2 | 28/37, 23 | 1 | +10.5 s |
| EQTransformer, causal, equal unmatched | 4/9, 2 | 20/37, 14 | 0 | +10.0 s |
| EQTransformer, causal, equal detections | 7/9, 6 | 27/37, 31 | 1 | +13.0 s |
| EQTransformer, look-ahead | 7/9, 1 | 28/37, 14 | 0 | (+12.5 s) |
| PhaseNet, causal | 7/9, 3 | 25/37, 12 | 0 | +9.5 s |
| PhaseNet, look-ahead, equal unmatched | 7/9, 5 | 31/37, 18 | 2 | (+12.5 s) |
| PhaseNet, look-ahead, equal detections | 6/9, 2 | 26/37, 12 | 0 | (+12.5 s) |

# S9 Software

**Table S9:** Software and hardware with which the results were produced.

| Component | Version | Used for |
|---|---|---|
| ayzek | v0.1.0-beta.3 | every pipeline result but FilterPicker |
| ayzek | v0.1.0-beta.5 | FilterPicker (the only change to the pipeline) |
| ayzek | v0.1.0-beta.6 | analysis tools only (pipeline as in beta.5) |
| ayzek | v1.0.0 (commit 3e43064) | the release of the tests: held-out tests, Pd magnitude, early location, retuned baselines |
| held-out tool | 440a2d9; 6fda3e2 | selection and fetch of the test sets (KO; New Zealand and Greece) |
| GCC | 16.2.1 | the x86 development build (glibc 2.44) |
| zig | clang 21.1, musl 1.2.5 | the static aarch64 build (Raspberry Pi) and the static x86-64 build of the tests (SHA-256 a40e7d23…a6899) |
| meson / ninja | 1.12.1 / 1.13.2 | build |
| qemu-aarch64 | 11.1 | the aarch64 test suite on x86 |
| Python | 3.12.13 | offline evaluation, training, continuous scan |
| PyTorch | 2.13.0 | offline evaluation and continuous scan |
| NumPy / SciPy | 2.5.1 / 1.18.0 | offline evaluation, verification |
| ObsPy | 1.5.0 | offline evaluation, verification |
| scikit-learn | 1.9.0 | event AUC |
| SeisBench | 0.12.6 (PyTorch 2.14.1, Python 3.13.16) | the published models |
| x86 machine | AMD Ryzen 5 5600 (6 cores, 12 threads), 31 GB, Arch Linux, kernel 7.2 | |
| GPU | NVIDIA RTX 3060 Ti, 8 GB (continuous scan and published models) | |
| Raspberry Pi | Raspberry Pi 4 Model B, Cortex-A72, 4 cores at 1.8 GHz | |

Reproduction: `tools/env_report.sh` prints the toolchain, the Python environments and the machine.

## S10 Single-window thresholds below the plateau

The adopted trigger rule fires on eight consecutive windows at 0.8 or one at 0.9. Table S10 compares it with a single-window rule at thresholds just below the 0.90–0.907 saturation plateau, with the instant branch off and chance corrected as in the main text. Placed just below the plateau, a single threshold performs comparably to the adopted rule. At 0.895 it detects 50.0 chance-corrected tuning events with 18 unmatched alarms and 33.6 Sındırgı $M_{\mathrm{w}}$ 6.1 events with 3, against 48.7 with 26 and 36.4 with 7. It is also sensitive to the third decimal, since at 0.899 the $M_{\mathrm{w}}$ 6.1 detections fall to 26. None of these thresholds was in the grid from which the adopted rule was selected.

**Table S10:** Single-window thresholds: detected / chance-corrected / unmatched alarms.

| Rule | Tuning (66 events) | Held out (66 events) |
|---|---|---|
| adopted ($0.8 \times 8$, or 0.9) | 51 / 48.7 / 26 | 39 / 36.4 / 7 |
| single window, 0.85 | 56 / 50.5 / 123 | 55 / 51.0 / 69 |
| single window, 0.87 | 53 / 47.0 / 89 | 52 / 48.8 / 39 |
| single window, 0.88 | 55 / 51.2 / 69 | 49 / 45.9 / 31 |
| single window, 0.89 | 55 / 52.6 / 41 | 41 / 38.1 / 11 |
| single window, 0.895 | 52 / 50.0 / 18 | 36 / 33.6 / 3 |
| single window, 0.899 | 45 / 42.3 / 13 | 26 / 24.1 / 2 |
| single window, 0.9 | 44 / 41.4 / 11 | 24 / 22.1 / 2 |

Reproduction: `tools/sweep_trigger.py --set 'threshold=0.85 0.87 0.88 0.89 0.895 0.899 0.9' --set trigger-windows=1 --set instant-threshold=2 --shifts 127,263,391` on the window probabilities of the three replays.

## S11 The quiet-window alarms, one by one

Table S11 lists the 16 alarms of the learned detector on the AFAD quiet windows with the class of Table S20b. The classes use two criteria that need no judgment of the waveforms. *KIRK pulse*: KIRK declared the alarm and its detection fell in the slot of its periodic pulse, 50–60 s after a 10-minute mark (218 of its 225 detections fall there, Figure S4c; in these nine alarms at 58.6–58.8 s); the alarm itself follows when the partner station's detection completes the pair, 62–105 s after the mark. *Seismic, picker-confirmed*: the rule below confirms a seismic source in at least one of its four runs. *Undetermined*: neither applies. For alarm 1 the pipeline's own picks suggest that an earthquake at BAND and CMH coincided with KIRK's pulse; no published picker confirms it, and the alarm is counted as a pulse.

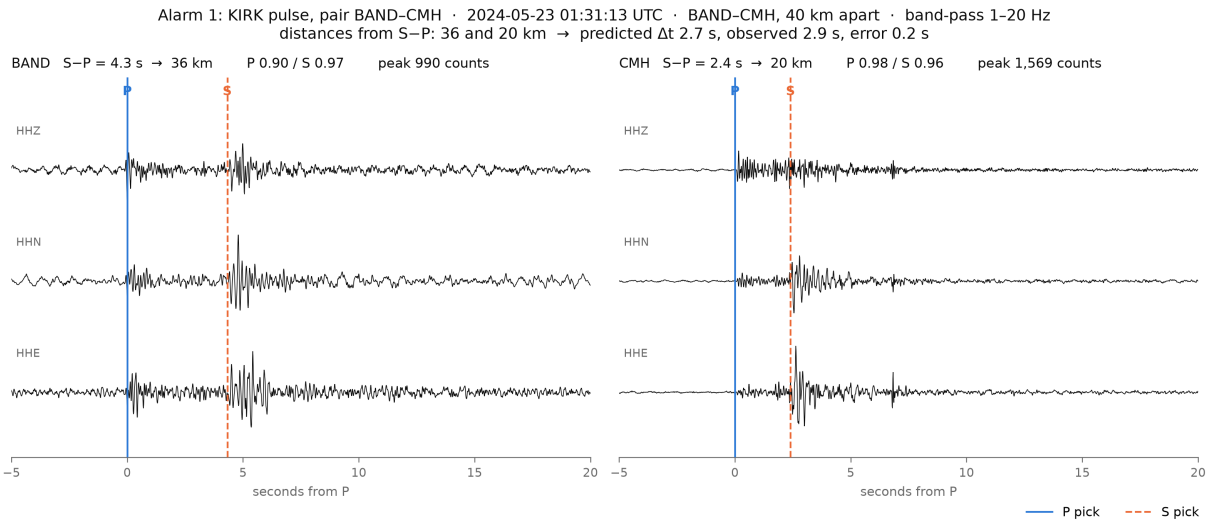
The Pickers column applies one rule, fixed before it was run, to every alarm of both detectors. A published picker must find P and S at the first two declaring stations at a probability of at least 0.5, each S−P interval gives a distance ($V_P = 6.0$, $V_S = 3.5$ km/s), and the two distances must fit the station separation and predict the observed difference of the two P picks to within 1.5 s. The four codes are PhaseNet with ETHZ and with STEAD weights, then EQTransformer with ETHZ and with STEAD weights (SeisBench 0.12.6; the INSTANCE weights, which are compared as detectors in the main text, are not used): C, confirmed; I, picks at both stations that fail the test; 1, a P and S pair at one station only; 0, at neither. No run confirms an alarm of KIRK's pulse. The same rule confirms three of the 323 STA/LTA alarms, the events of alarms 13, 14 and 15.

The figures show every station pair that detected within the coincidence window of one of the alarms and has the pipeline's P and S picks at both stations (19 pairs; alarm 11 has none), titled by the alarm and its class; the picks are shown for orientation and the time in each title is the P pick at the first station, not an origin time. After them comes a random sample (seed 20261022) of 30 of the 323 STA/LTA alarms, of which 19 have P and S picks to mark. Band-pass 1–20 Hz; P picks solid, S picks dashed; the three components of a panel share one amplitude scale. The distances in the titles follow from the S−P intervals of the picks whether or not the picks lie on seismic arrivals; for an alarm without a seismic source they have no physical meaning.

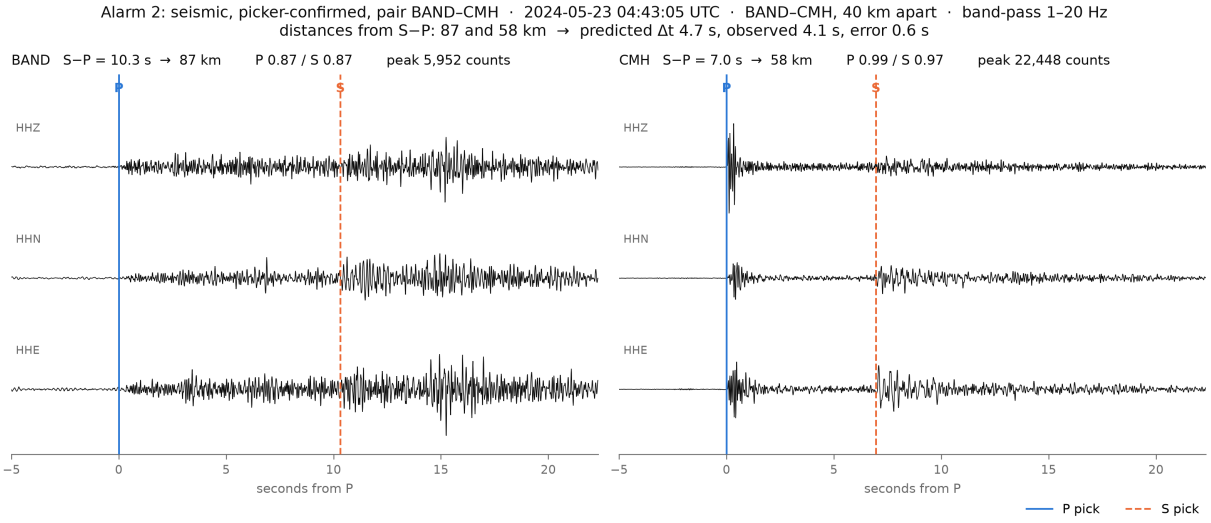
**Sensitivity analyses.** Two analyses, specified before they were run (Table S13), test how conservative the classification is. Applied to every pair of stations associated with an alarm (the declaring stations and any station whose detection is compatible with theirs under the pipeline's association rule) rather than to the first two declaring stations, the rule also confirms alarm 1 in three of its four runs, through the BAND–CMH pair, as a local event that coincided with KIRK's pulse. It confirms no other pulse alarm and leaves alarms 4, 8 and 16 undetermined. For STA/LTA it confirms six alarms, two of them on events the learned detector did not alarm on. Without the confirmed alarms the rates are 2.57 and 74.0 per day. KOERI's FDSN service has no event service, so the EMSC catalog and the ISC bulletin, which carries KOERI's own locations, were searched for every alarm (origin up to 180 s before it, within 300 km). The only events that fit an alarm are the M2.4 of alarm 2 (EMSC and KOERI) and an M1.9 located by KOERI for alarm 13. Neither catalog lists any event for the undetermined alarms 4, 8 and 16, or for any pulse alarm. KIRK's pulse does not appear in the event replays. One of 38 KIRK detections on the Marmara replay falls in the pulse slot, in no unmatched alarm, and none of 20 on the $M_w$ 6.1 replay, and every pulse alarm on the quiet windows falls between 23 May and 9 June 2024.
Reproduction: `tools/classify_alarms.py --pairs any` and `tools/alarm_catalogues.py`.
Reproduction: the classes are in `figures/supp/quiet_alarms.csv` of the paper's repository; the panels are drawn by `tools/plot_candidates.py --titles figures/supp/quiet_alarm_panels.csv` on the misfire table of `tools/misfire_review.py`.

**Table S11:** The 16 alarms of the learned detector on the AFAD quiet windows.

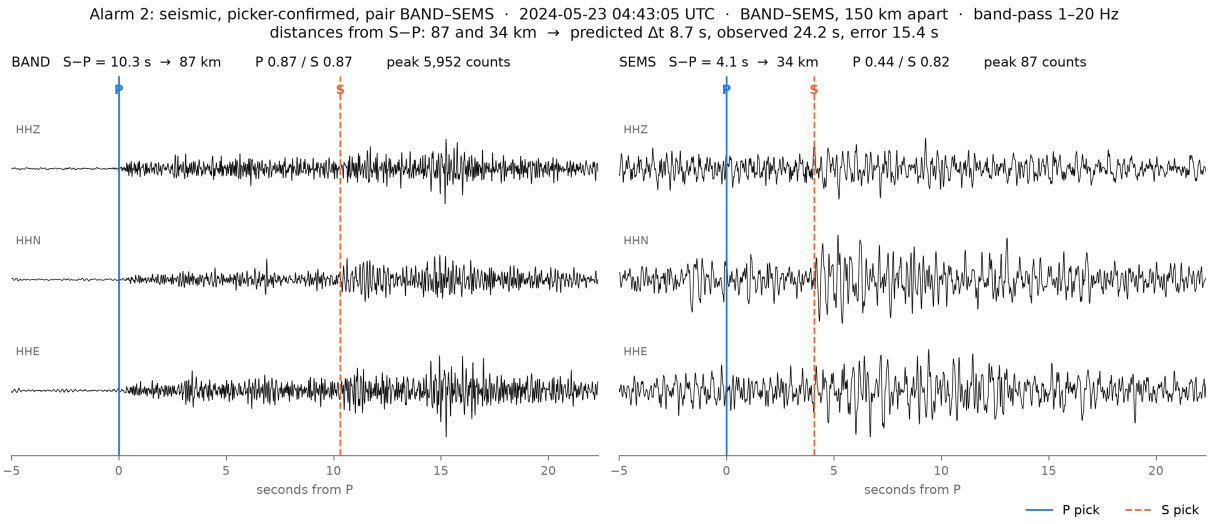
| # | Date | UTC | Declared by | Class | Pickers | Evidence |
|---|---|---|---|---|---|---|
| 1 | 2024-05-23 | 01:31:11 | CMH, KIRK | KIRK pulse | `1 1 1 1` | KIRK's detection 58.6 s past a 10-minute mark; the pipeline's own picks also suggest an earthquake at BAND and CMH |
| 2 | 2024-05-23 | 04:43:07 | BAND, CMH | seismic, picker-confirmed | `1 1 C C` | confirmed by EQTransformer ETHZ, EQTransformer STEAD; EMSC M2.4, origin 04:42:52, 53/78 km from CMH/BAND |
| 3 | 2024-05-23 | 22:41:02 | KIRK, MANT | KIRK pulse | `0 0 0 1` | KIRK's detection 58.7 s past a 10-minute mark |
| 4 | 2024-05-24 | 00:58:59 | BAND, CMH | undetermined | `0 0 1 1` | no P and S pair at both stations in any picker run |
| 5 | 2024-05-29 | 10:01:35 | KIRK, SEMS | KIRK pulse | `1 1 1 1` | KIRK's detection 58.7 s past a 10-minute mark |
| 6 | 2024-05-30 | 18:01:24 | BAND, KIRK | KIRK pulse | `0 0 0 1` | KIRK's detection 58.6 s past a 10-minute mark |
| 7 | 2024-06-03 | 06:31:33 | BAND, KIRK | KIRK pulse | `0 0 0 0` | KIRK's detection 58.6 s past a 10-minute mark |
| 8 | 2024-06-03 | 06:44:17 | BAND, SEMS | undetermined | `0 1 0 1` | no P and S pair at both stations in any picker run |
| 9 | 2024-06-05 | 03:21:29 | BAND, KIRK | KIRK pulse | `0 0 0 0` | KIRK's detection 58.6 s past a 10-minute mark |
| 10 | 2024-06-05 | 11:01:45 | KIRK, SEMS | KIRK pulse | `1 1 0 1` | KIRK's detection 58.8 s past a 10-minute mark |
| 11 | 2024-06-09 | 12:51:02 | KAND, KIRK | KIRK pulse | `0 0 0 0` | KIRK's detection 58.6 s past a 10-minute mark |
| 12 | 2024-06-09 | 13:51:03 | KAND, KIRK | KIRK pulse | `0 0 0 0` | KIRK's detection 58.6 s past a 10-minute mark |
| 13 | 2025-02-26 | 17:18:01 | BAND, CMH | seismic, picker-confirmed | `C C 1 C` | confirmed by PhaseNet ETHZ, PhaseNet STEAD, EQTransformer STEAD |
| 14 | 2025-02-26 | 17:28:08 | BAND, CMH | seismic, picker-confirmed | `1 1 C C` | confirmed by EQTransformer ETHZ, EQTransformer STEAD |
| 15 | 2025-02-27 | 12:48:41 | KIRK, VIZE | seismic, picker-confirmed | `0 0 0 C` | confirmed by EQTransformer STEAD; KIRK's detection is outside the pulse slot |
| 16 | 2025-03-02 | 15:30:20 | BAND, CMH | undetermined | `1 1 I 1` | picks at both stations fail the S−P test in one run; no pair in the others |



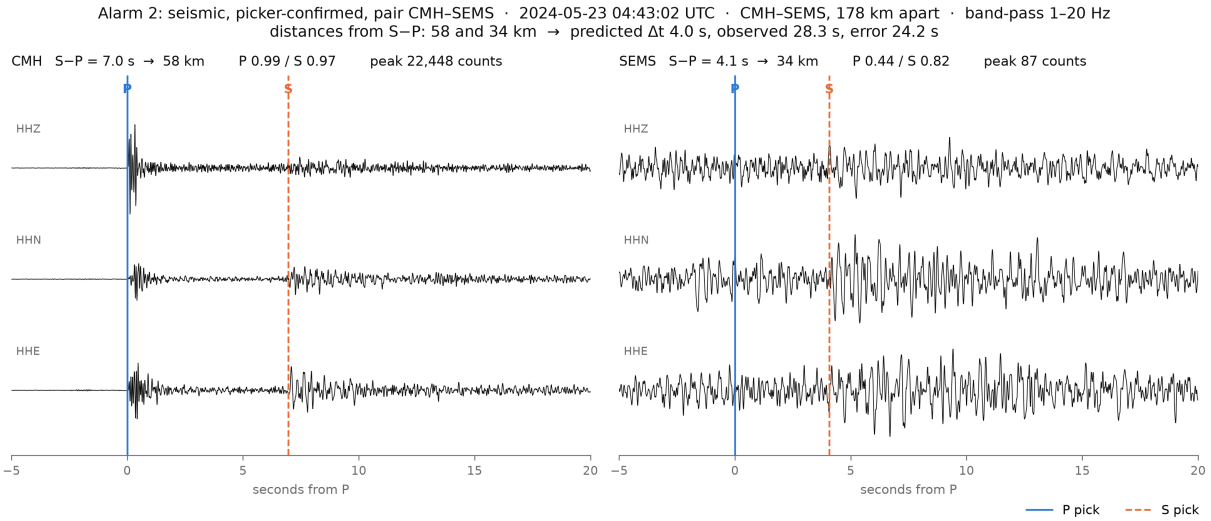

Alarm 1: KIRK pulse, pair BAND–CMH.



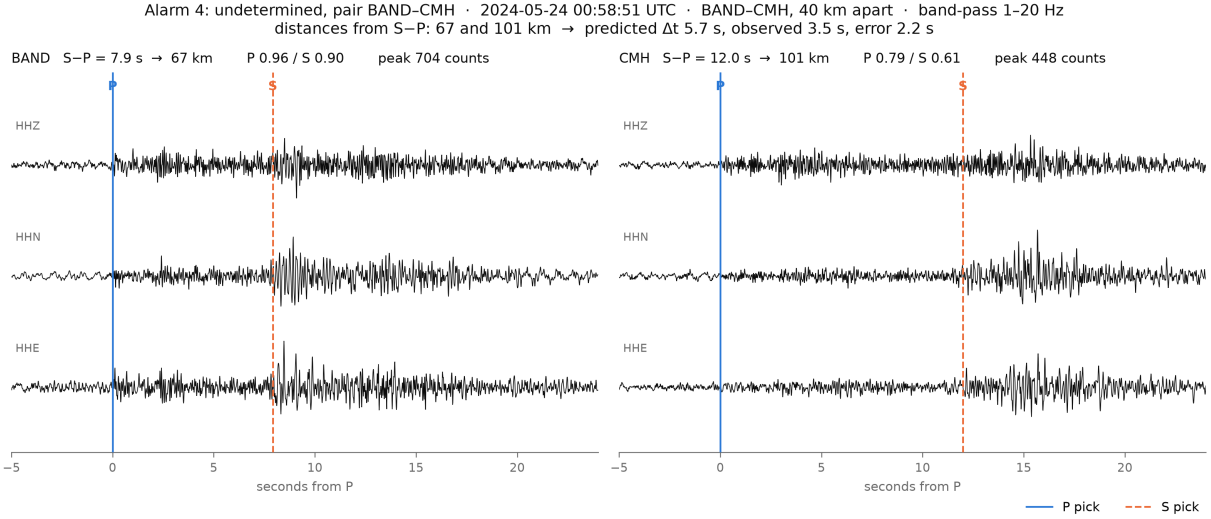

Alarm 2: seismic, picker-confirmed, pair BAND–CMH.



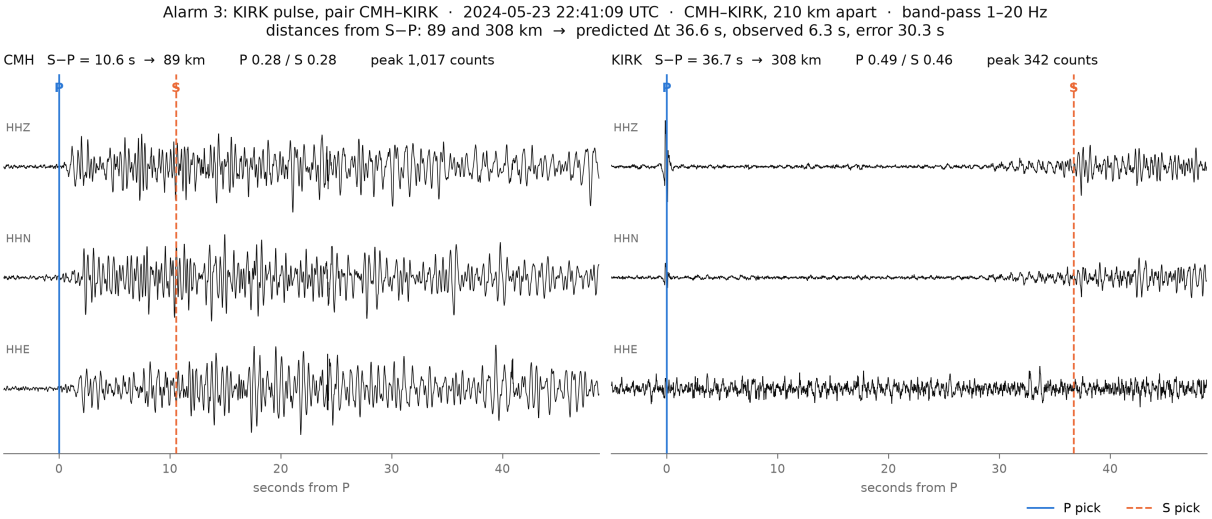

Alarm 2: seismic, picker-confirmed, pair BAND–SEMS.



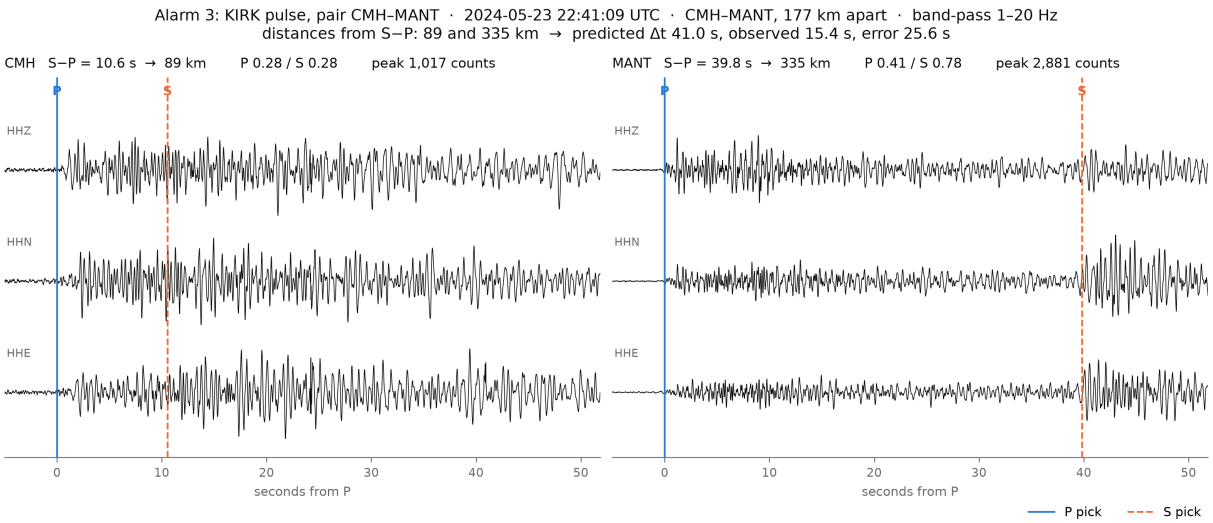

Alarm 2: seismic, picker-confirmed, pair CMH–SEMS.



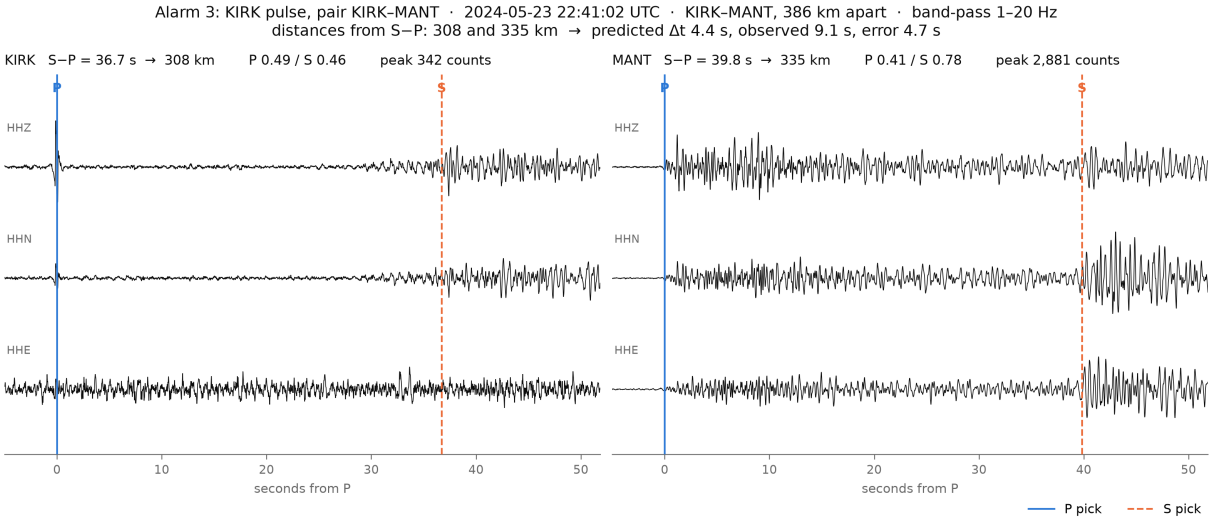

Alarm 4: undetermined, pair BAND–CMH.



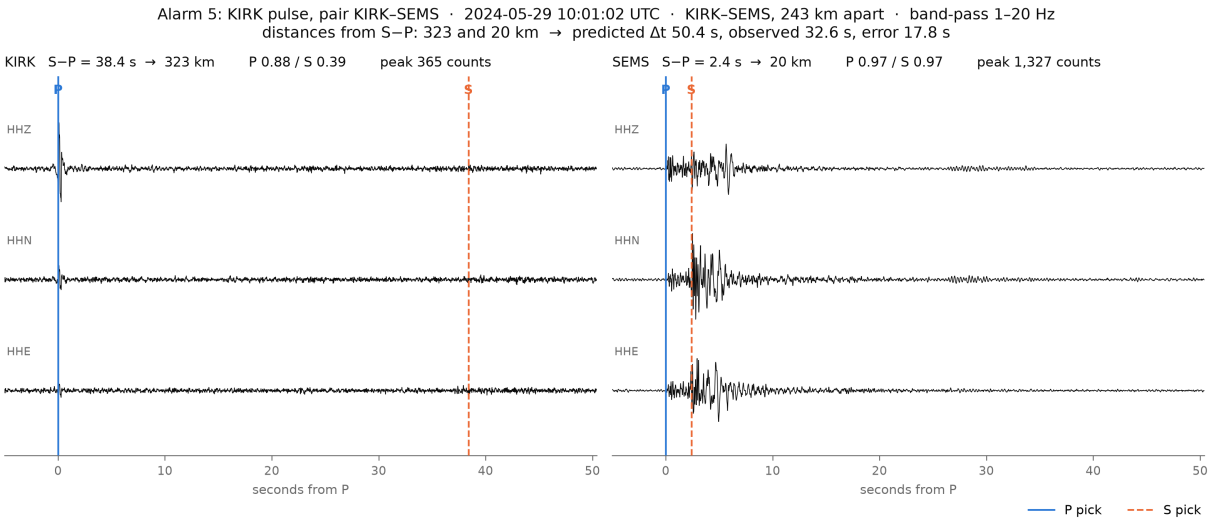

Alarm 3: KIRK pulse, pair CMH–KIRK.



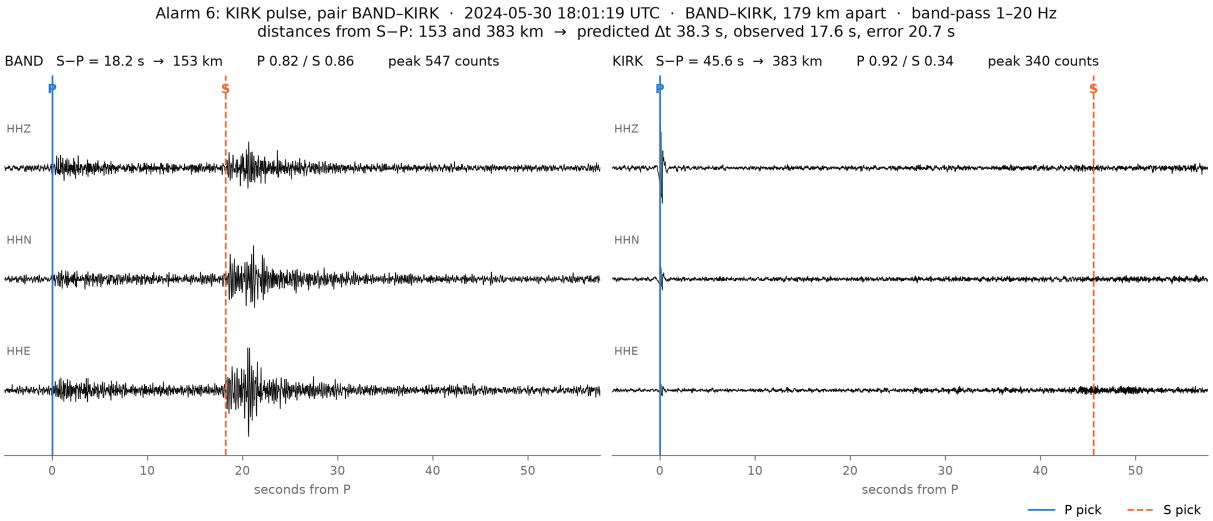

Alarm 3: KIRK pulse, pair CMH–MANT.



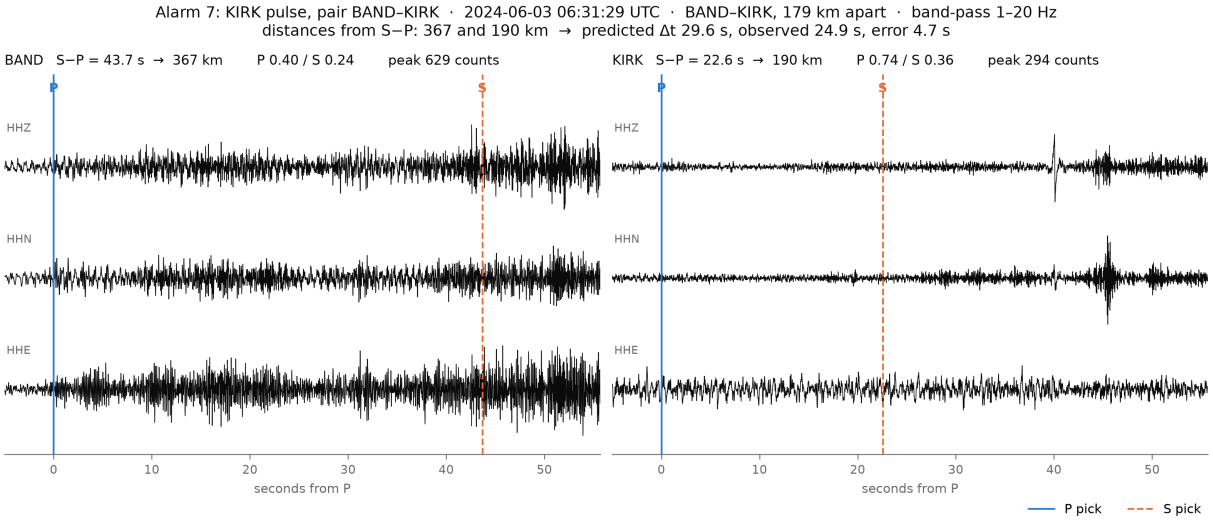

Alarm 3: KIRK pulse, pair KIRK–MANT.



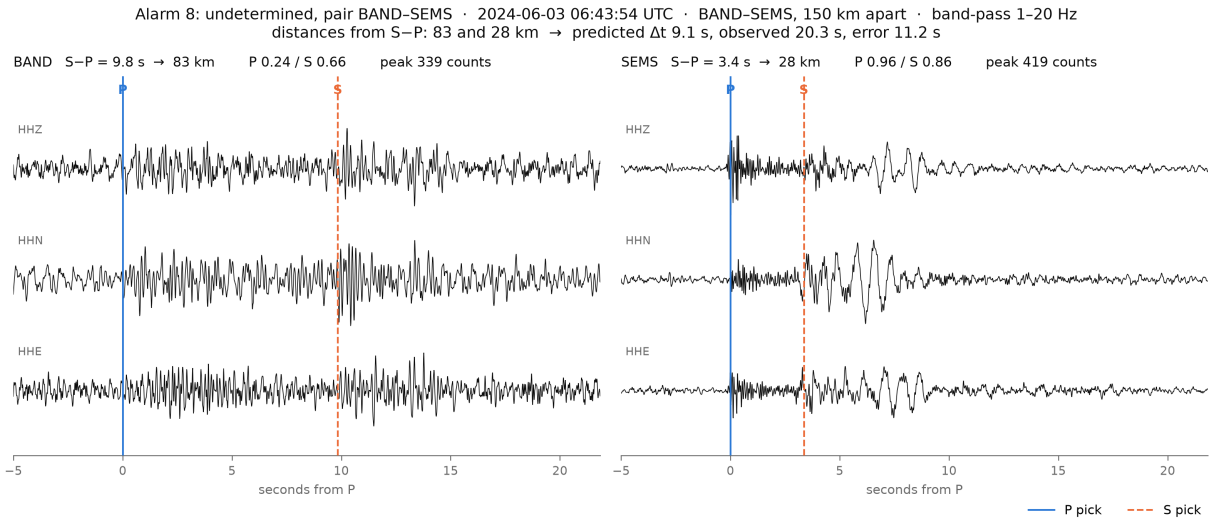

Alarm 5: KIRK pulse, pair KIRK–SEMS.



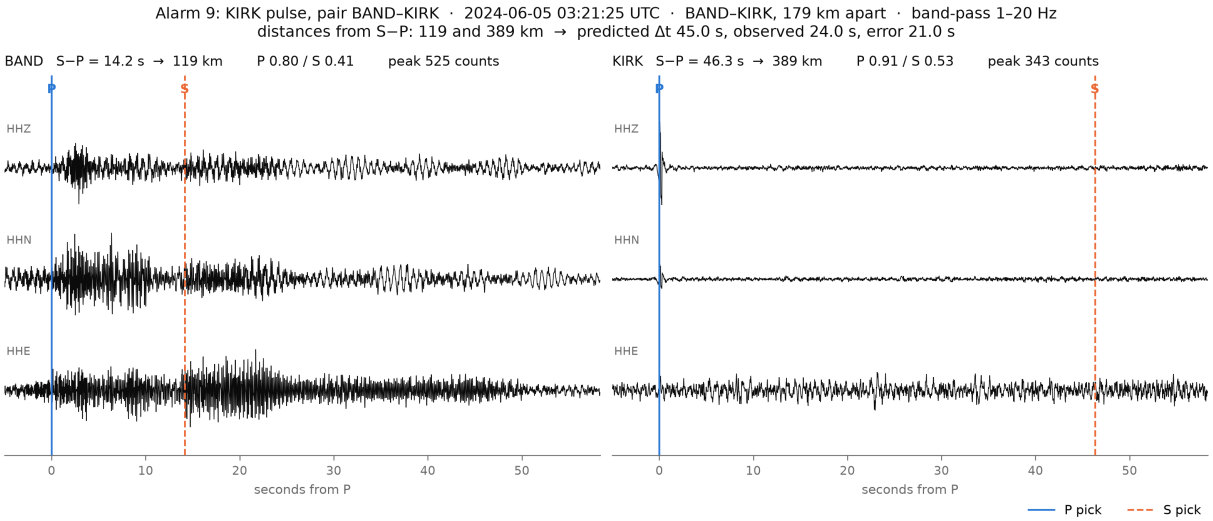

Alarm 6: KIRK pulse, pair BAND–KIRK.



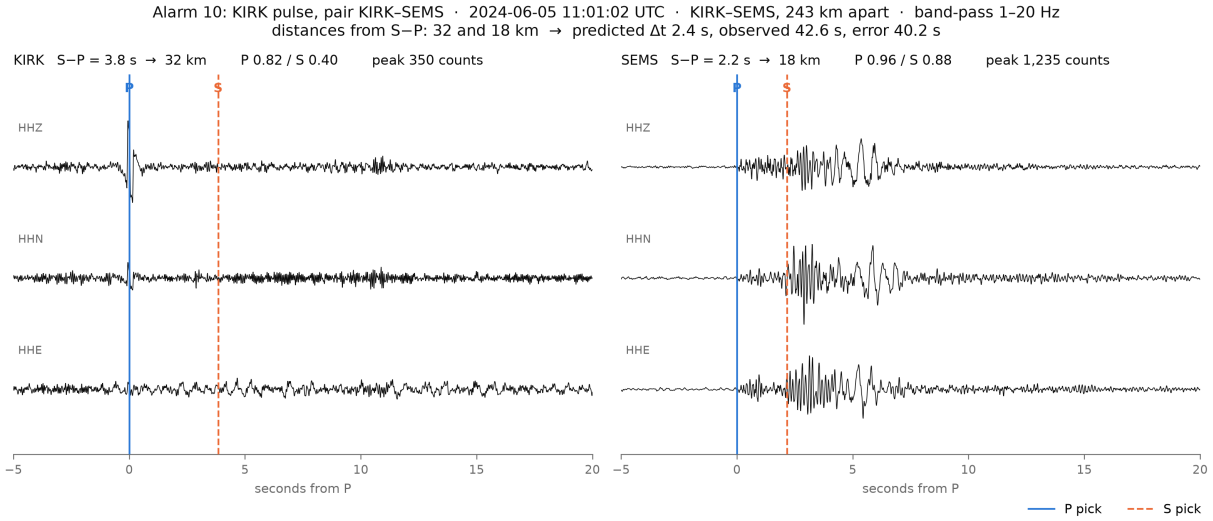

Alarm 7: KIRK pulse, pair BAND–KIRK.



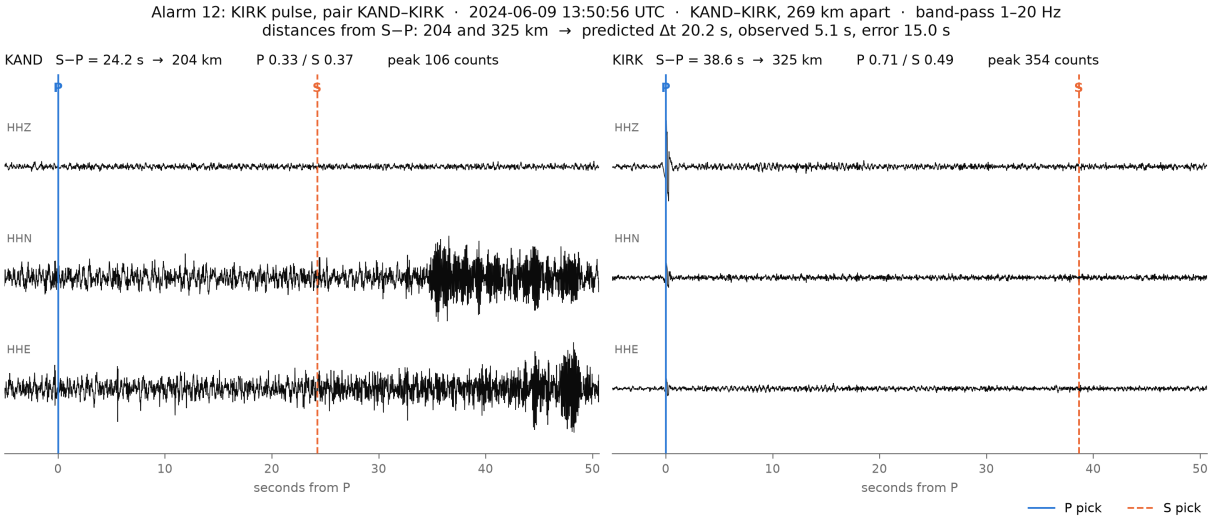

Alarm 8: undetermined, pair BAND–SEMS.



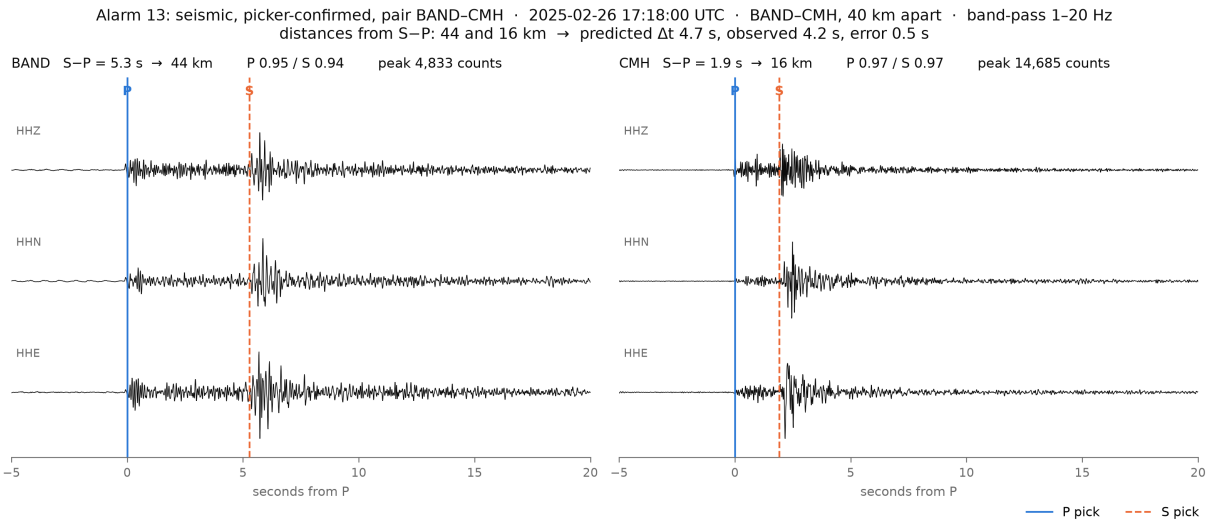

Alarm 9: KIRK pulse, pair BAND–KIRK.



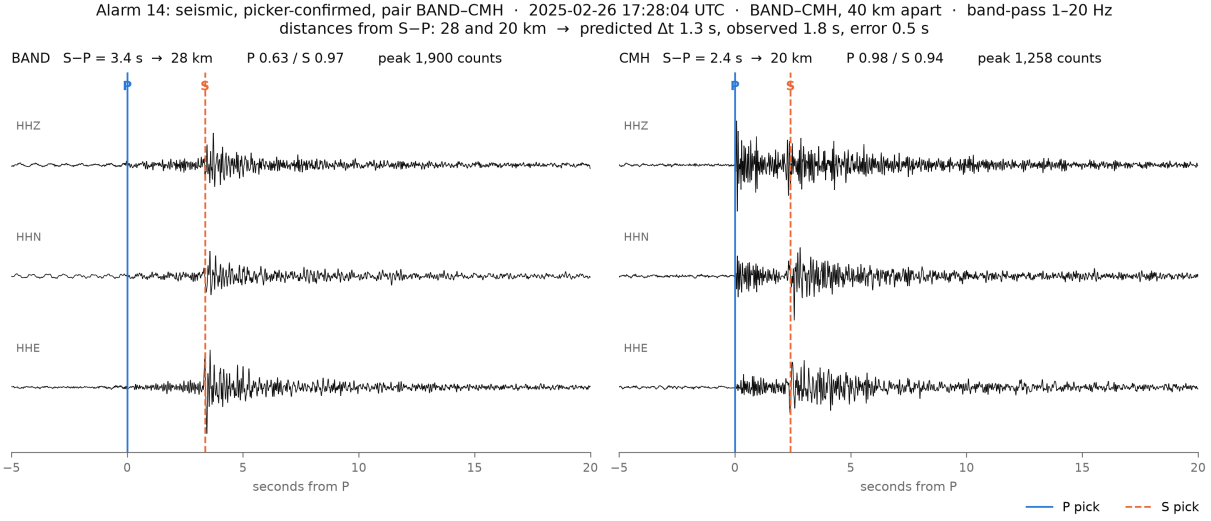

Alarm 10: KIRK pulse, pair KIRK–SEMS.



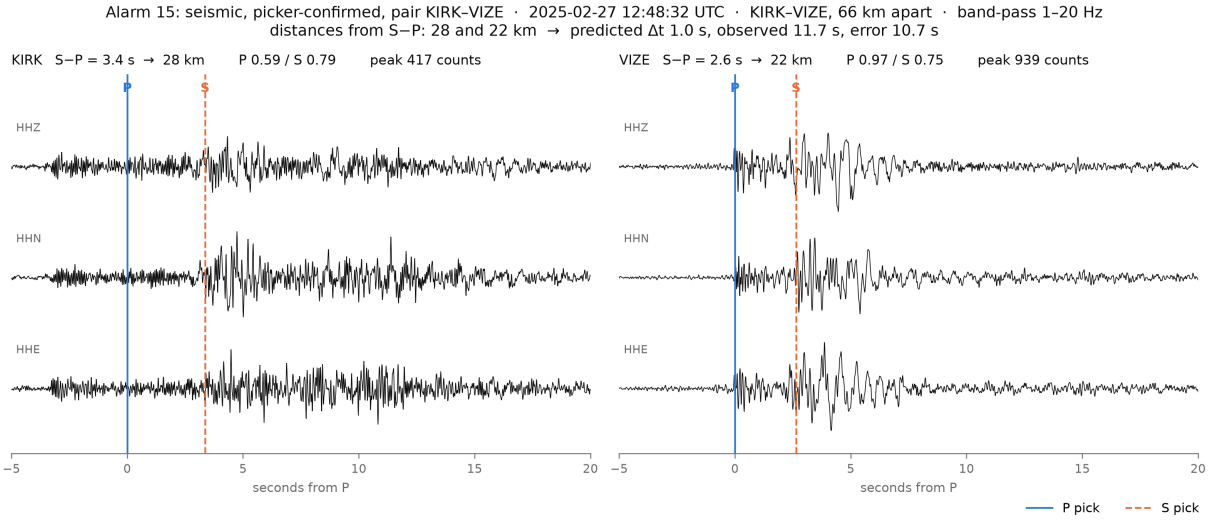

Alarm 12: KIRK pulse, pair KAND–KIRK.



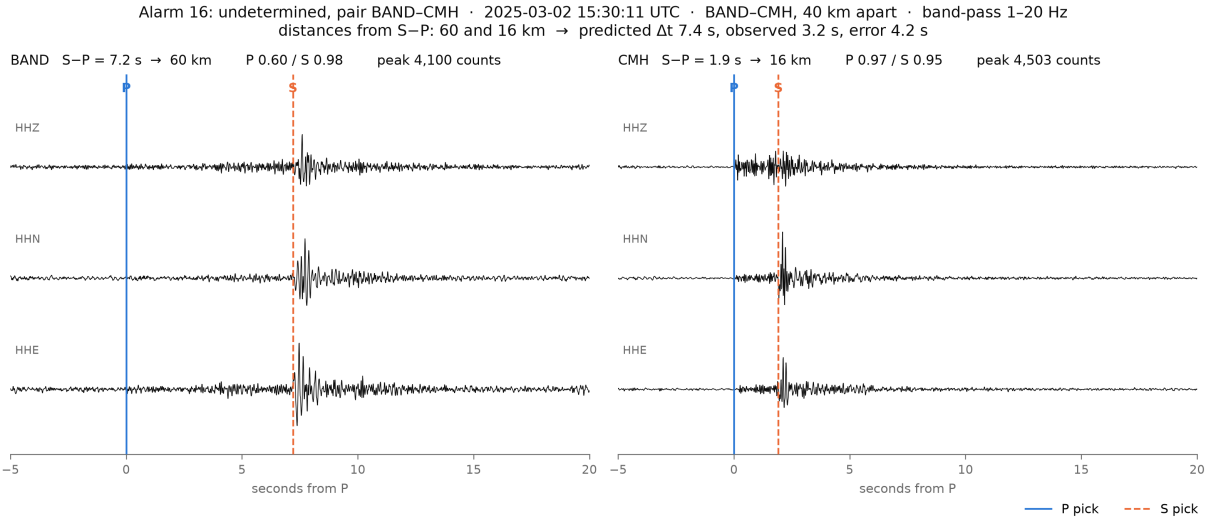

Alarm 13: seismic, picker-confirmed, pair BAND–CMH.



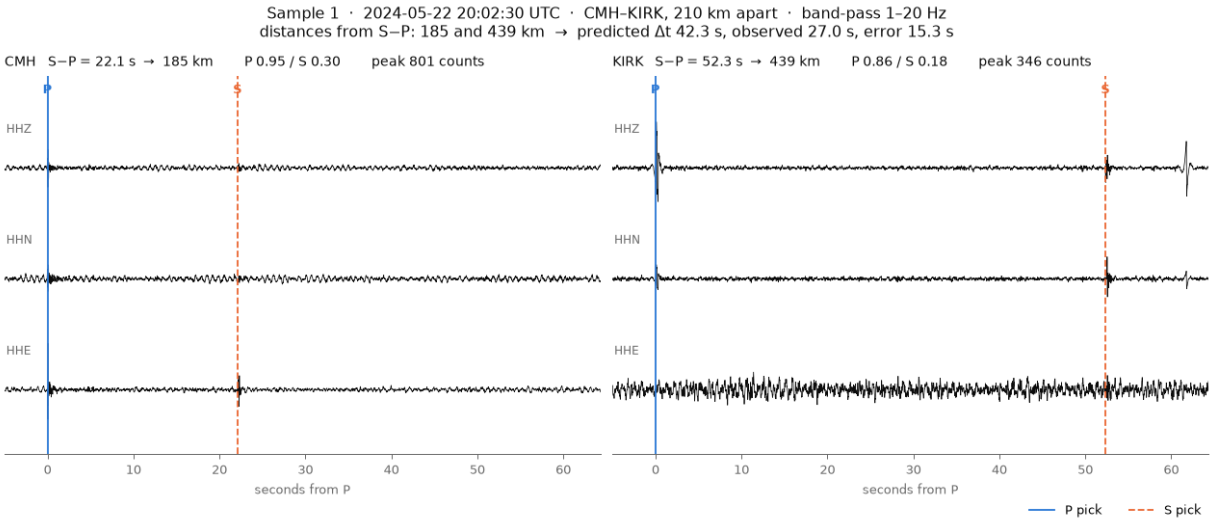

Alarm 14: seismic, picker-confirmed, pair BAND–CMH.



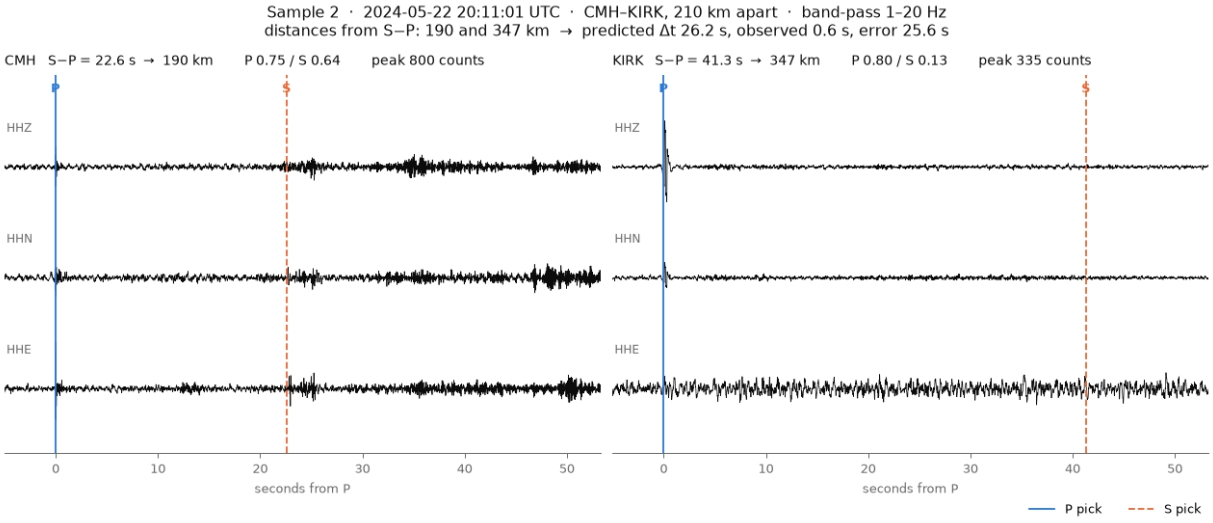

Alarm 15: seismic, picker-confirmed, pair KIRK–VIZE.



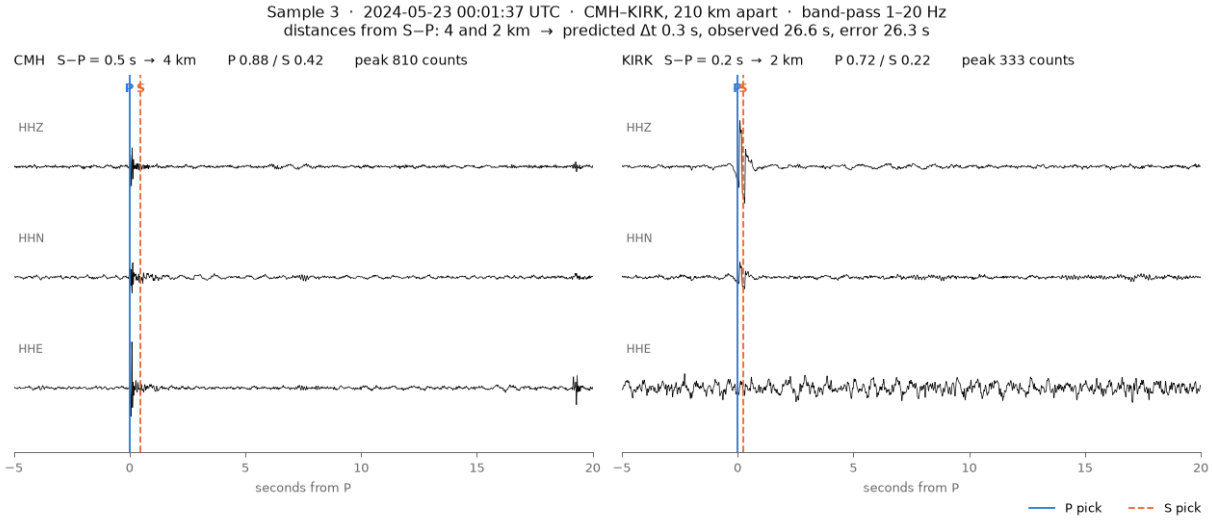

Alarm 16: undetermined, pair BAND–CMH.



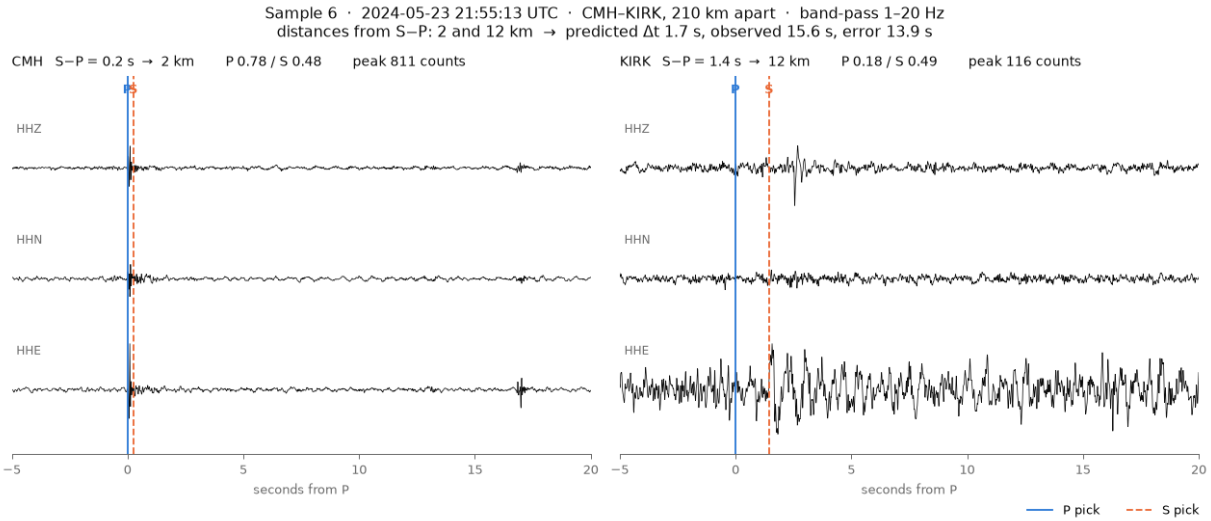

STA/LTA; window 20240522 1924; CMH–KIRK; random sample.



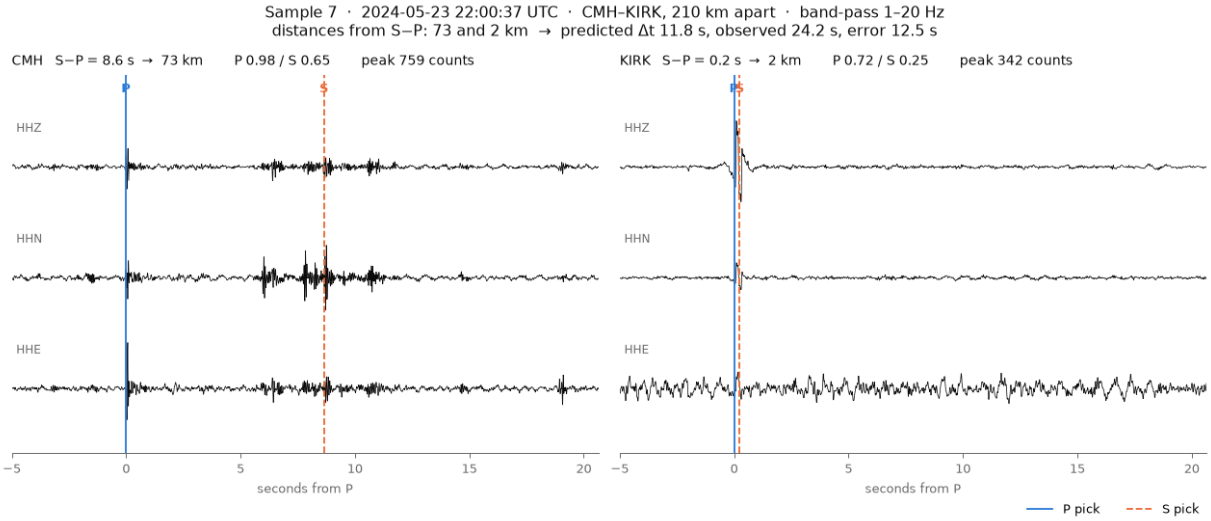

STA/LTA; window 20240522 1924; CMH–KIRK; random sample.



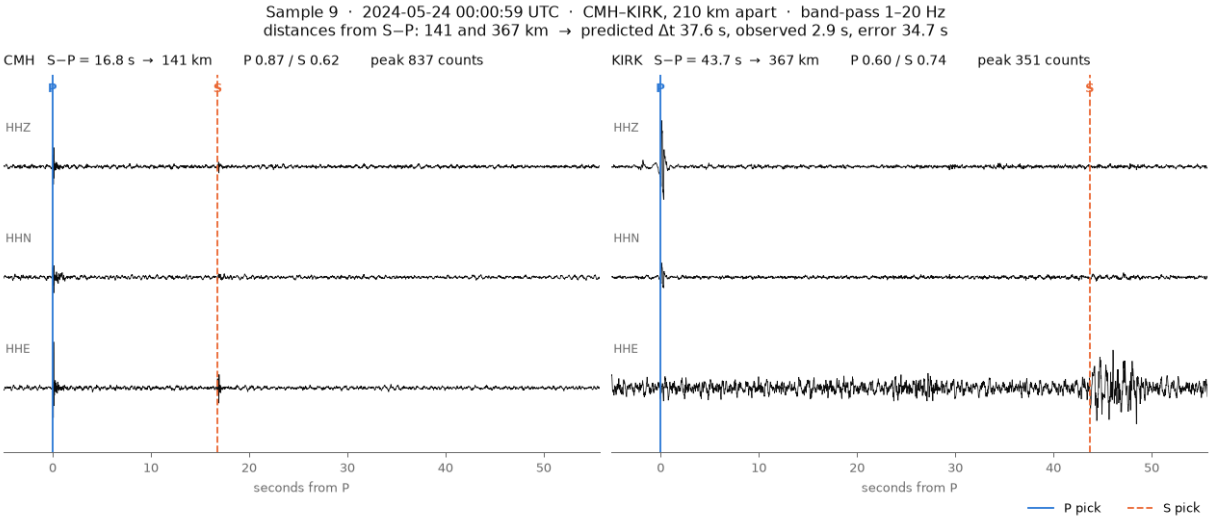

STA/LTA; window 20240522 1924; CMH–KIRK; random sample.



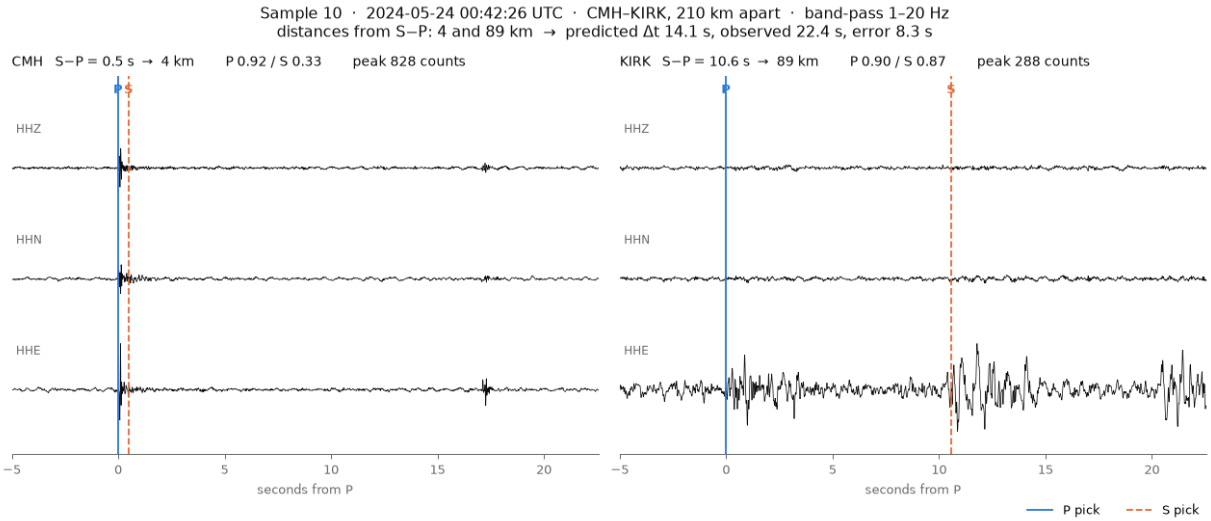

STA/LTA; window 20240523 2121; CMH–KIRK; random sample.



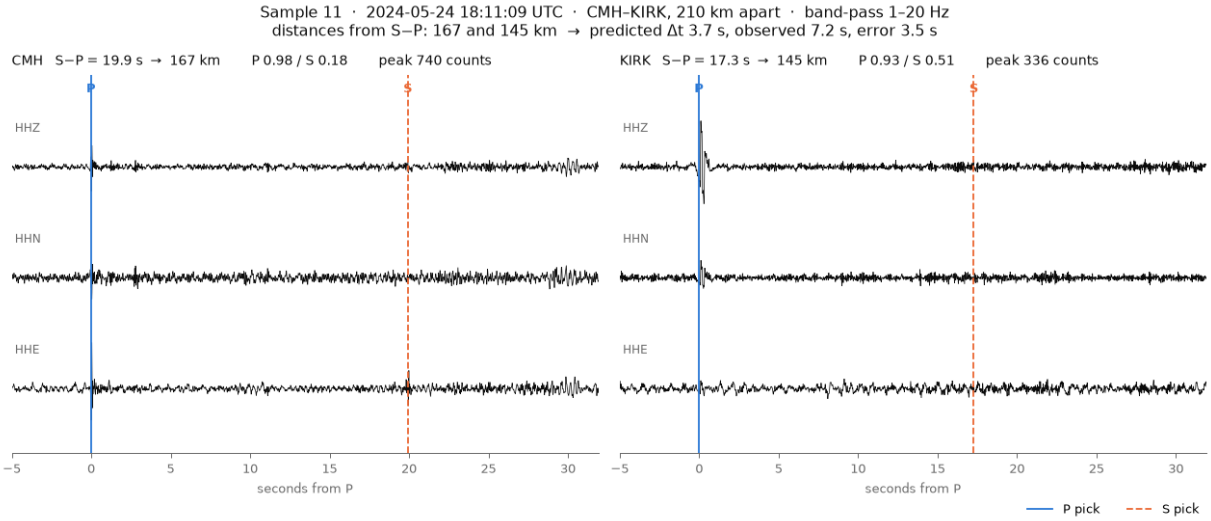

STA/LTA; window 20240523 2121; CMH–KIRK; random sample.



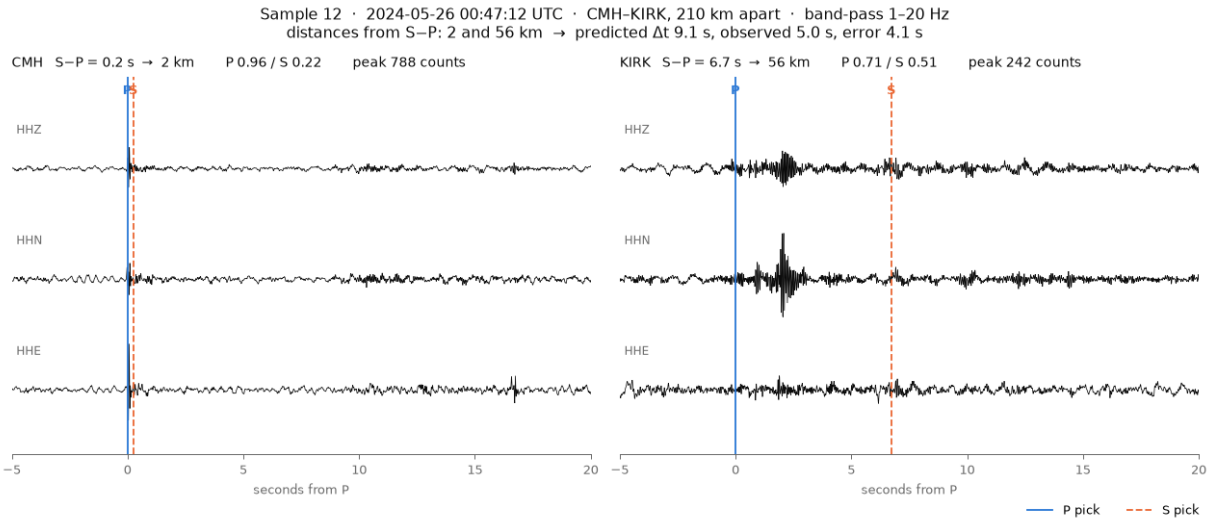

STA/LTA; window 20240523 2121; CMH–KIRK; random sample.



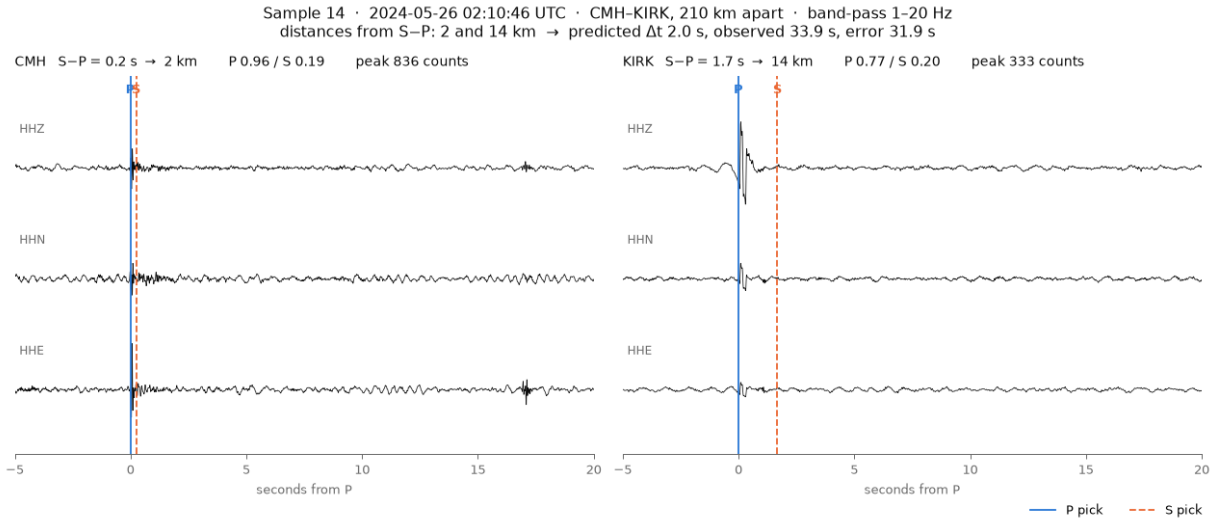

STA/LTA; window 20240523 2121; CMH–KIRK; random sample.



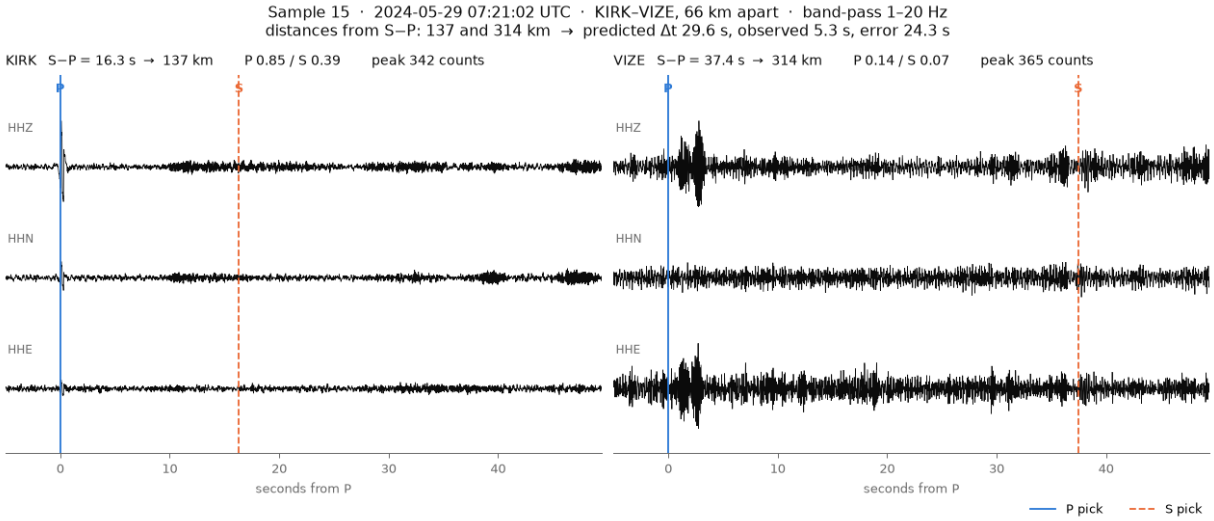

STA/LTA; window 20240524 1450; CMH–KIRK; random sample.



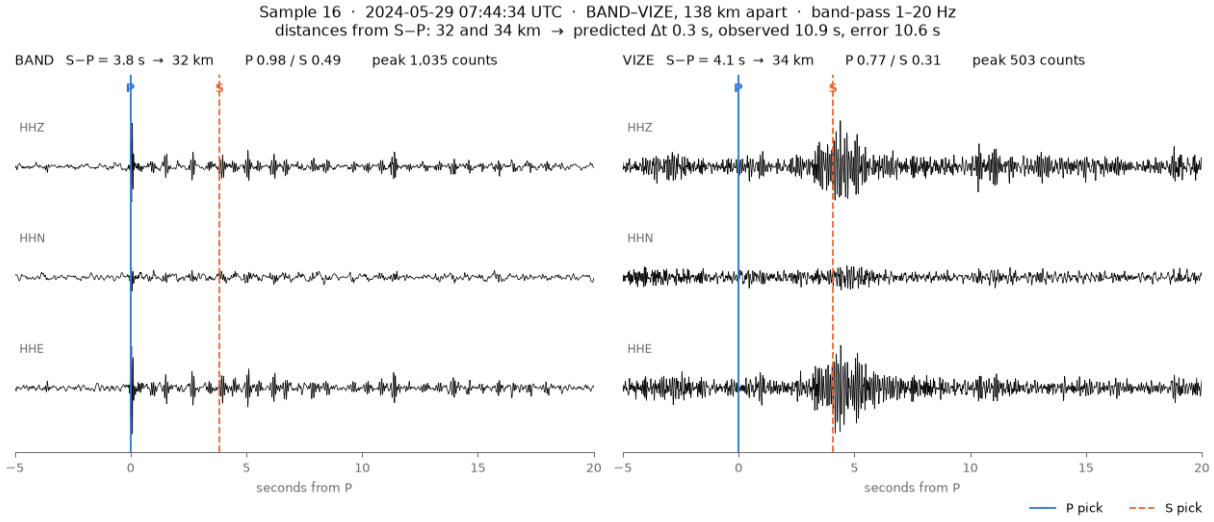

STA/LTA; window 20240526 0001; CMH–KIRK; random sample.



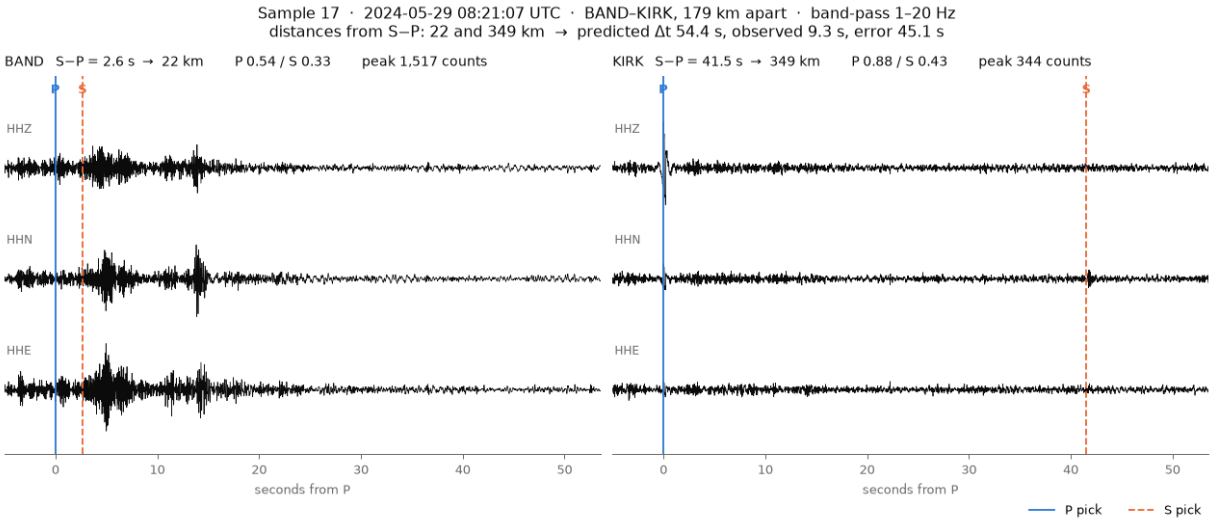

STA/LTA; window 20240526 0001; CMH–KIRK; random sample.



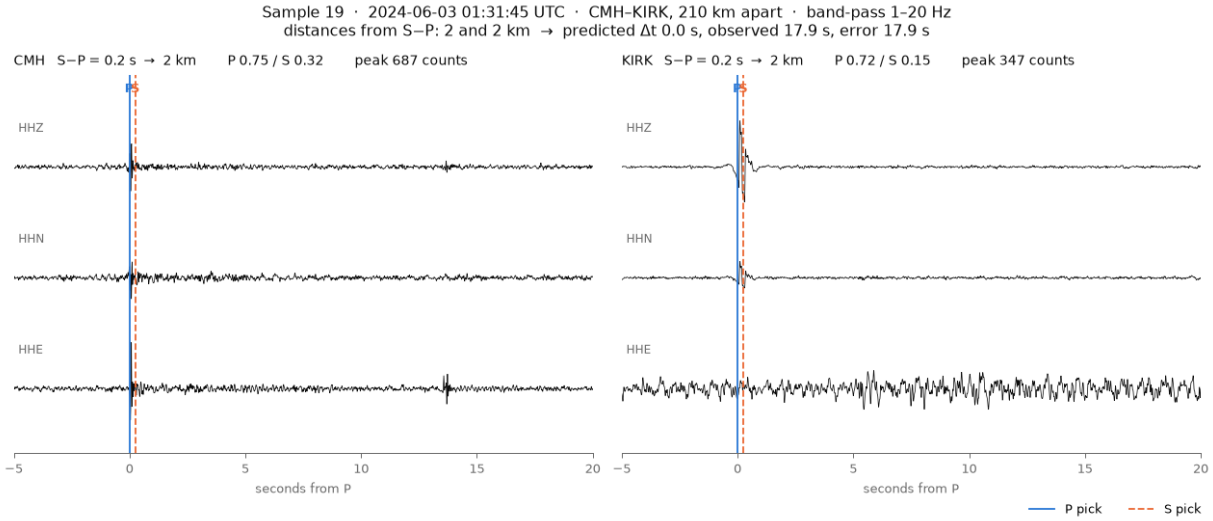

STA/LTA; window 20240529 0638; KIRK–VIZE; random sample.



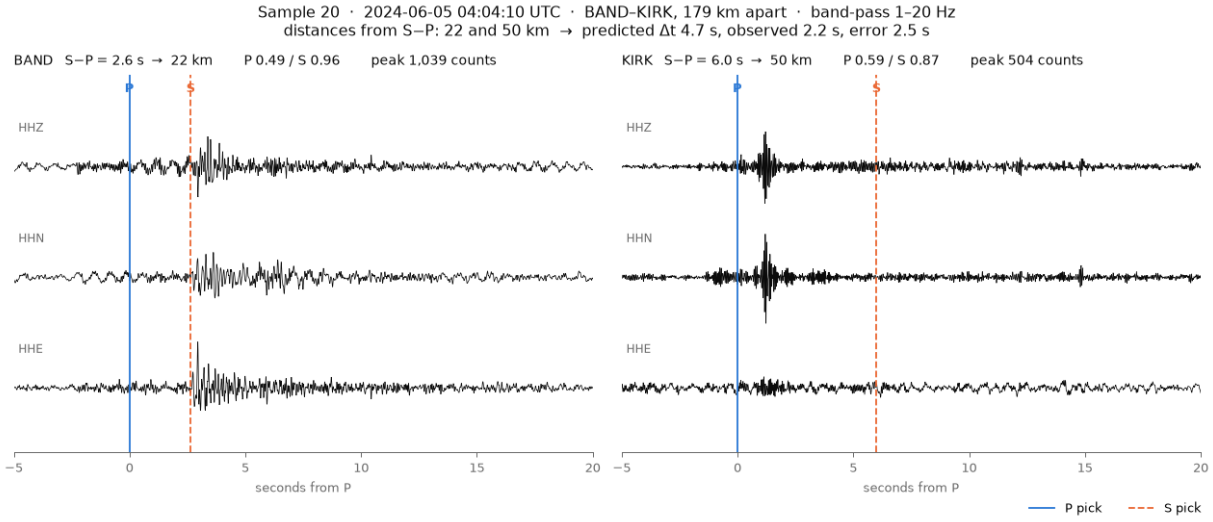

STA/LTA; window 20240529 0638; BAND–VIZE; random sample.



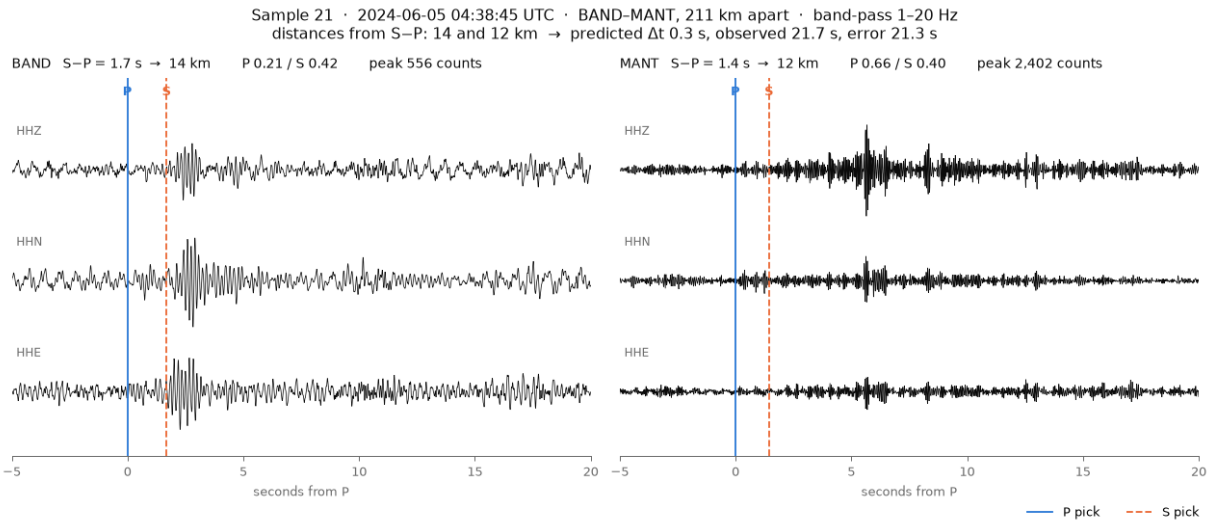

STA/LTA; window 20240529 0638; BAND–KIRK; random sample.



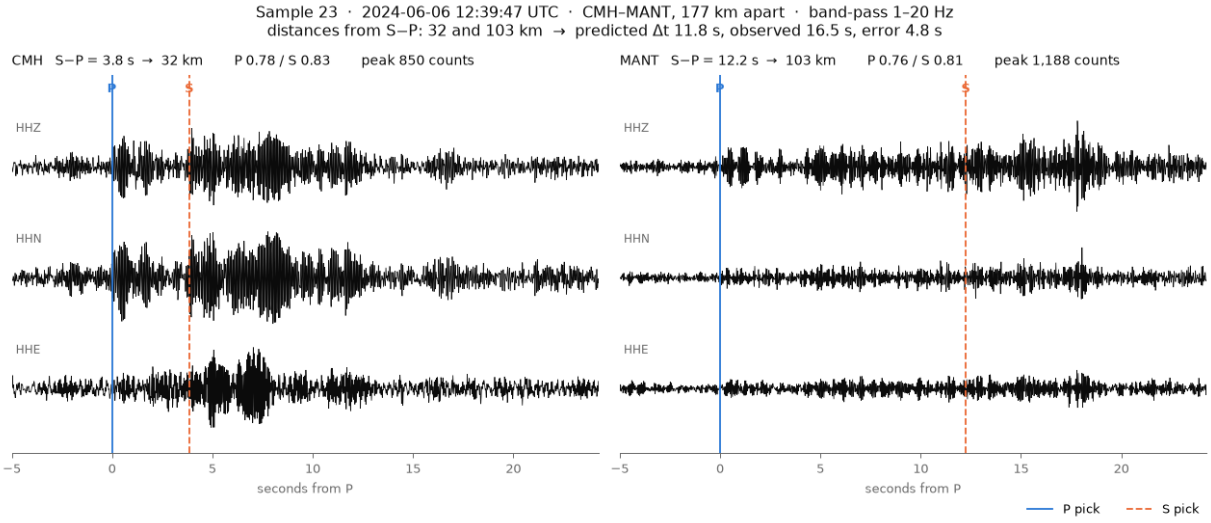

STA/LTA; window 20240602 2300; CMH–KIRK; random sample.



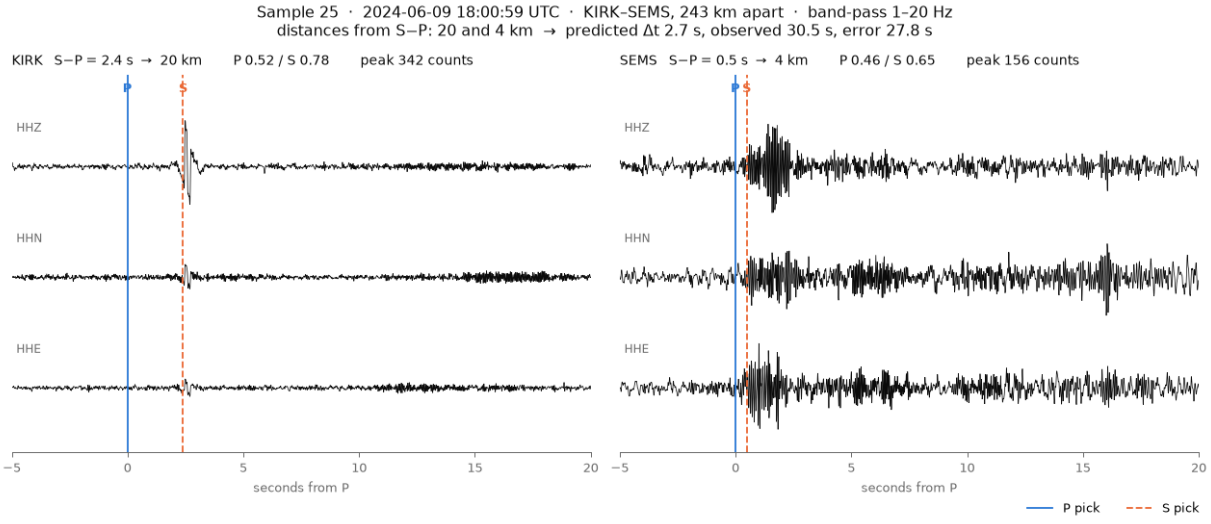

STA/LTA; window 20240605 0140; BAND–KIRK; random sample.



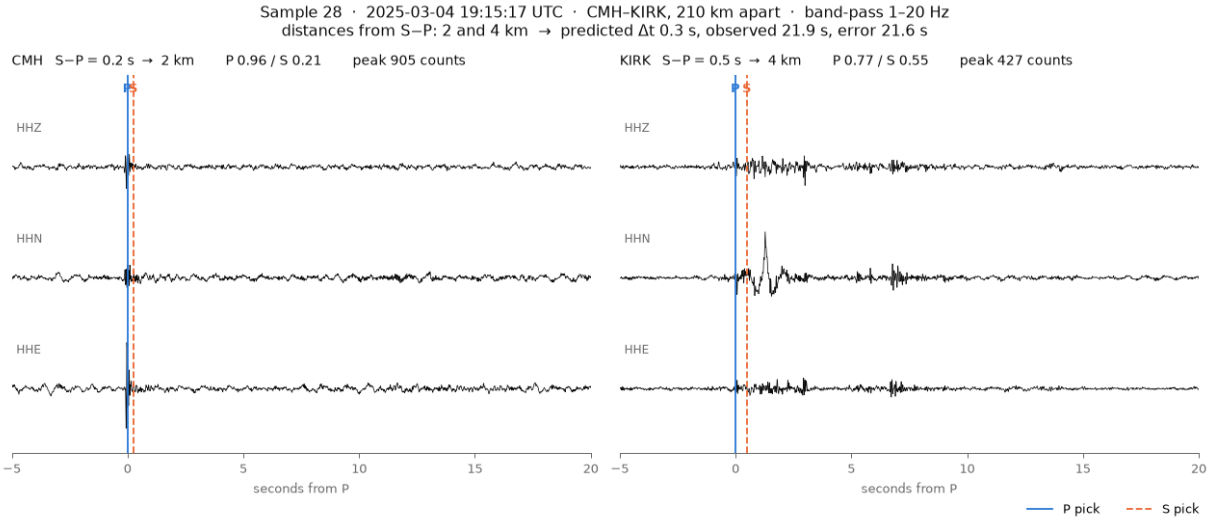

STA/LTA; window 20240605 0140; BAND–MANT; random sample.




STA/LTA; window 20240606 1024; CMH–MANT; random sample.




STA/LTA; window 20240609 1210; KIRK–SEMS; random sample.




STA/LTA; window 20250304 1830; CMH–KIRK; random sample.

## S12 FilterPicker: verification, grid and selection

The pipeline's FilterPicker is a port of Lomax's FilterPicker5 v5.0.3 (C). The test `tests/test_filterpicker.cpp` builds Lomax's code unchanged and compares the two on synthetic data and on 4 min of DEMI under three parameter sets, and every pick and every sample of the characteristic function agree. The detector runs on the vertical component; every pick triggers, at most once per 15 s per station, with P at the pick time and no STA/LTA anchoring, and everything downstream (picker, association with two stations and a 3 s slack, catalog scoring) is the pipeline's.

The protocol was committed before the tuning grid was run (Section S13). Two runs preceded it as checks of the port, both on tuning data at the published defaults. They replayed the AFAD Sındırgı half hour on all seven stations of its data directory, and the Marmara replay. The grid on the tuning replays crossed two sets of windows (filter, long-term and $T_{\mathrm{up}}$: 3, 5 and 0.2 s, the values of Lomax et al. (2012) at 100 Hz; 6, 10 and 0.4 s, those of Earthworm's `pick_FP` at 100 Hz) with trigger thresholds of 3, 4, 5, 6, 8, 10, 12, 15 and 20 and acceptance thresholds of 3, 4, 6, 8, 10, 15 and 20 (126 settings), each scored at the catalog times and the six shifts. The two rules of the main text were applied against the learned detector's tuning figures (48.7 chance-corrected detections, 26 unmatched alarms); ties went to fewer unmatched alarms and then to higher thresholds (equal unmatched), or to more detections (equal detections).

**Table S12:** FilterPicker on the tuning replays (Sındırgı 30 min and Marmara, 66 catalog events).

| Setting | Windows (s) | Thresholds | Detected | Chance-corrected | Unmatched |
|---|---|---|---|---|---|
| equal unmatched (selected) | 6 / 10 / 0.4 | 15, 10 | 46 | 42.3 | 25 |
| equal detections (selected) | 3 / 5 / 0.2 | 4, 10 | 53 | 49.8 | 52 |
| published defaults | 3 / 5 / 0.2 | 10, 10 | | 47.5 | 78 |
| published defaults | 6 / 10 / 0.4 | 10, 10 | | 41.0 | 30 |
| learned detector | $0.8 \times 8$, or 0.9 | | 51 | 48.7 | 26 |

Over the grid, the most chance-corrected detections at a given number of unmatched alarms are 36.4 at 4, 37.3 at 6, 39.6 at 20, 42.3 at 25, 44.8 at 41 and 49.8 at 52. Both selected settings alarm on the Sındırgı $M_{\mathrm{w}}$ 4.9 (+16.6 and +17.3 s after origin) and the Marmara $M_{\mathrm{w}}$ 6.2 (+9.9 s). On the quiet windows the stations of the equal-unmatched setting's 353 alarms are CMH (300), KIRK (274), VIZE (45), SEMS (28), MANT (25), BAND (22), ELBA (11) and KAND (1).
Reproduction: `tools/sweep_trigger.py` with `--args '--detector filterpicker'`, `--set fp-windows=...` and `--set fp-thresholds=...`; `tools/select_setting.py` applies the rules; `tools/event_outcomes.py` and `tools/significance.py` give the paired tests of the main text.

## S13 Record of protocols committed in advance

Every choice that fixed a setting or an analysis before the development sequences or quiet data were scored, and the steps of the release and the tests, with the commit that records it. Commits in *ayzek* are in the pipeline's repository, the others in the paper's. Times are commit times (UTC+3), recorded by the authors' own repositories and not timestamped by a third party; the order of steps therefore rests on these records.

The evaluation runs followed the protocols. The published models were evaluated in fba5ffb (2026-10-22 19:16) and the Simav sequence in 49fe24f (2026-10-23 13:01). FilterPicker was evaluated on 2026-10-26 between 21:15 and 21:20, with its results in c06566f, and the look-ahead settings on 2026-10-27 between 22:02 and 23:16, with their results in 40003af. No setting was changed after any evaluation run. The commits from the release on are timestamped by GitHub and, for the release, by Zenodo (10.5281/zenodo.23806743).

**Table S13:** The steps committed in advance, their records and what preceded each.

| Step | Commit | Date | Run before it |
|---|---|---|---|
| learned detector: trigger rule chosen on the tuning replays | *ayzek* 500254a | 2026-09-17 02:51 | the tuning grid; the first held-out result is in the same commit, and the order therefore rests on the authors' record |
| STA/LTA: the two selection rules | *ayzek* a04dc9c | 2026-09-17 20:49 | the tuning grid; the held-out results are in the same commit |
| published models: protocol | 91b9ee1 | 2026-10-22 16:27 | one adapter check on the Sındırgı 30 min tuning replay, recorded in 571ef5f |
| published models: selection | 4973cb8 | 2026-10-22 17:54 | the tuning grid; no held-out, Simav or quiet-window run |
| second held-out sequence (Simav): protocol | 51067a1 | 2026-10-23 09:49 | a list of candidate sequences by catalog counts and station distances; no waveform, score or replay |
| quiet-window alarms: classification rule, PhaseNet | 0a0dff8 | 2026-10-23 10:27 | no picker run |
| quiet-window alarms: EQTransformer cross-check | 8380650 | 2026-10-23 10:36 | the PhaseNet run of the rule above |
| FilterPicker: protocol | 851f3fb | 2026-10-26 20:43 | two runs at the published defaults on tuning data (Section S12) |
| FilterPicker: selection | 9580e0f | 2026-10-26 21:15 | the tuning grid; no held-out, Simav or quiet-window run |
| quiet-window alarms: any-pair rule and catalog search | c78517b | 2026-10-27 18:24 | no run of either |
| published models, 1–2 s look-ahead: protocol | 761d22b | 2026-10-27 21:13 | one check that `--ahead 0` reproduces the causal scores |
| published models, 1–2 s look-ahead: selection | a6bdd22 | 2026-10-27 21:58 | the tuning grid; no held-out, Simav or quiet-window run |
| *The release and the tests (UTC)* | | | |
| baselines retuned on the five development replays (three rules) | *ayzek* (protocol, section 6) | 2026-10-29 | every development replay; no test pool touched |
| protocol fixed; release v1.0.0 tagged, pushed and deposited on Zenodo | *ayzek* 3e43064 | 2026-10-30 12:55:36 | no waveform from any test pool fetched |
| KO selection written | held-out tool 440a2d9 | 2026-10-30 13:05 | the KO catalog and availability requests only |
| New Zealand selection written | held-out tool 440a2d9 (stamped 6fda3e2) | 2026-10-30 23:04 | GeoNet's catalog and day-file listing only |
| Greece selection written | held-out tool 6fda3e2 | 2026-10-31 06:42 | NOA's catalog and availability requests; four earlier attempts died in the availability stage |
| KO test run (`tools/run_heldout.py`, static v1.0.0 binary) | v1.0.0 | 2026-10-31 10:55 to 14:00 | the fetched KO sets, verified against their request records; the analysis stage was rerun at 14:01 with `tools/network_subsets` rebuilt from the tagged source (the static copy predated its `--quiet` option), no replay repeated |
| New Zealand test run | v1.0.0 | 2026-10-31 14:00 to 20:22 | the fetched New Zealand sets, verified |
| Greece test run (on a second machine, same binary by SHA-256) | v1.0.0 | 2026-10-31 19:52 to 2026-11-01 02:48 | the fetched Greek sets, verified; the published models' window scores from an RTX 3050 on the second machine, all replays and the analysis on the first (the replays not needing published-model scores ran while the scores were computed), instead of the RTX 3060 Ti of the other two runs (checked on one KO set: 2 of 151 116 |

## S14 Detector ranking on a continuous record

Six detectors were scored on a continuous 728-day record at the development station MANT: the learned detector and the pipeline's STA/LTA, and four offline detectors, a classical STA/LTA (Allen, 1978), two P-wave detectors that score 3.4 s windows and differ only in their noise windows, and a 6 s detector with station scaling (Table S14a, Figure S1). Windows from 10 s before to 60 s after the P arrival of any cataloged event within 500 km form its guard and the rest the background; the threshold for a budget of $B$ window false alarms per day is the score that $B$ background windows a day exceed, and an event is detected when the highest score in its guard exceeds it. The pipeline's two detectors are almost indistinguishable by event AUC (0.9910 for STA/LTA, 0.9892 for the learned detector; paired difference on the common events 0.0016, 0.0000–0.0035), yet at one window false alarm per day the learned detector finds 34.8% of the events against 22.2%, 1.6 times as many, and among the offline detectors recall at that budget is in the exact reverse order of event AUC. The inversion is a property of the metric, which integrates over all false positive rates whereas operation uses rates of order $10^{-4}$.

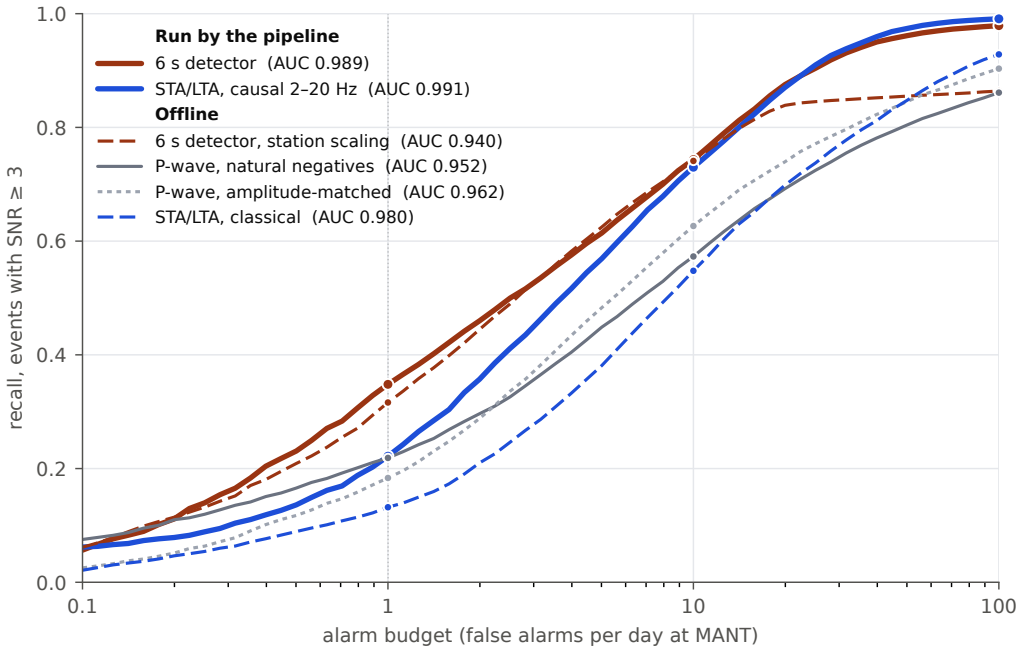



**Figure S1:** Recall of the six detectors of Table S14a against the alarm budget (window false alarms per day), 20 budgets a decade; dots mark the budgets of the table. Solid wide lines: detectors run by the pipeline; dashed: their offline counterparts.

**Without the S wave.** Panel (a) scores an event by the highest window score in its guard, from 10 s before the predicted P (minus the window length) to 60 s after it, and every detector can therefore score the S wave and coda as well as the P wave. Here an event is scored only from windows that end at least a margin before its predicted S arrival (IASP91), and only events whose S−P interval is at least 6 s plus the margin are counted, so that every detector, including the 6 s detectors on their 6 s grid, has a window ending between P and that limit. Thresholds are those of panel (a), because they are set by the background, which does not depend on how events are scored. The event AUC compares each event's maximum with the maximum over one random stretch of record of the same length, clear of every cataloged event.

At one window false alarm per day the ranking by recall is the reverse of the ranking by event AUC with and without the S wave, for the pipeline's pair and for the four offline detectors. Without the S wave the learned detectors lose most, and at ten window false alarms per day

**Table S14:** Recall at three alarm budgets (window false alarms per day at the one station) and event AUC on the 728-day MANT record.

**(a)** The 13,056 events with SNR $\geq 3$. Event AUC: maximum score within each event's guard window against the maximum over random stretches of equal length, away from cataloged events, of the same record.

| Detector | Event AUC | 100/day | 10/day | 1/day |
|---|---|---|---|---|
| *Run by the pipeline* | | | | |
| STA/LTA (causal, 2–20 Hz) | 0.9910 | 0.991 | 0.730 | 0.222 |
| learned detector | 0.9892 | 0.979 | 0.743 | 0.348 |
| *Offline* | | | | |
| STA/LTA (classical) | 0.9795 | 0.928 | 0.548 | 0.132 |
| P-wave, amplitude-matched | 0.9622 | 0.903 | 0.627 | 0.183 |
| P-wave, natural negatives | 0.9516 | 0.861 | 0.573 | 0.219 |
| 6 s detector, station scaling | 0.9403 | 0.864 | 0.741 | 0.316 |

**(b)** Each event scored over its full guard and only before its S wave (margin 1 s; 12,206 events with SNR $\geq 3$ and S$-$P $\geq 7$ s).

| | Full guard | | | | Before the S wave | | | |
|---|---|---|---|---|---|---|---|---|
| Detector | AUC | 100/day | 10/day | 1/day | AUC | 100/day | 10/day | 1/day |
| *Run by the pipeline* | | | | | | | | |
| STA/LTA (causal, 2–20 Hz) | 0.9911 | 0.991 | 0.724 | 0.217 | 0.9889 | 0.966 | 0.688 | 0.214 |
| learned detector | 0.9896 | 0.980 | 0.753 | 0.350 | 0.9765 | 0.899 | 0.649 | 0.338 |
| *Offline* | | | | | | | | |
| STA/LTA (classical) | 0.9784 | 0.927 | 0.546 | 0.130 | 0.9830 | 0.886 | 0.531 | 0.127 |
| P-wave, amplitude-matched | 0.9611 | 0.904 | 0.626 | 0.181 | 0.9271 | 0.766 | 0.461 | 0.162 |
| P-wave, natural negatives | 0.9501 | 0.862 | 0.573 | 0.218 | 0.9099 | 0.704 | 0.433 | 0.195 |
| 6 s detector (station scaling) | 0.9428 | 0.866 | 0.741 | 0.317 | 0.8691 | 0.670 | 0.521 | 0.247 |

STA/LTA draws level or ahead in both pairs. Margins of 0 s (12,566 events) and 2 s (11,944 events) give the same rankings at one window false alarm per day; the pipeline's detector then finds 0.339 and 0.336 of the events against 0.213 and 0.211 for its STA/LTA. Over the full guard, the values on this subset differ from panel (a), which counts all 13,056 events, by at most 0.010 in recall and 0.003 in AUC.

The event AUC also depends on which events are counted. Over all 48,432 cataloged events within 500 km of MANT in the scanned record, whatever their SNR, the event AUC of the offline 6 s detector is 0.675; it rises to 0.912, 0.940 and 0.960 when only events with an SNR of at least 2, 3 and 5 are counted. Panel (a) uses the 13,056 events with SNR $\geq 3$, from a complete rescan of the 728-day record (79 million window scores, 10.5 million for each 6 s detector and 18.5 million for each 3.4 s detector).

Reproduction: `tools/s_wave_control.py --runs <cnn_earthquake runs_mant> --margins 0 1 2`; it first reproduces the thresholds and full-guard recalls of panel (a) and stops if they differ.

## S15 Published models with a short look-ahead

In causal mode a published model scores each window at the very end of its input. To test whether that edge explains the gap between the causal and the centered modes, the input was made to end 1 or 2 s after the window (EQTransformer and PhaseNet, INSTANCE weights), a live system then scoring each window that much later. The protocol, the grid and the two rules are those of Section S8, committed before any of these scores existed; the settings were selected on the tuning replays and committed before the one evaluation run (Table S13). Alarm times include the look-ahead.

**Table S15:** Published models with 1 or 2 s of input past the window, beside the causal (0 s) and centered settings of Table S20a.

| Model, look-ahead | Rule, setting | Held out | | Simav | | Quiet | $M_w$ 6.1 |
|---|---|---|---|---|---|---|---|
| | | corr./unm. | difference | corr./unm. | difference | alarms | alarm |
| learned detector | | 36.4 / 7 | | 35.1 / 17 | | 16 | +17.0 s |
| PhaseNet, 0 s | both, 0.7 × 2 | 33.3 / 12 | +3.1 (−4.0, 10.5) | 36.8 / 28 | −1.8 (−8.2, 4.1) | 15 | +16.5 s |
| PhaseNet, 1 s | e.u., 0.6 × 4 | 35.4 / 11 | +1.0 (−5.6, 7.6) | 36.0 / 25 | −0.9 (−6.7, 4.8) | 17 | +18.0 s |
| PhaseNet, 1 s | e.d., 0.7 × 4 | 30.0 / 8 | +6.4 (−0.7, 13.7) | 36.2 / 20 | −1.1 (−6.7, 4.5) | 10 | +18.5 s |
| PhaseNet, 2 s | both, 0.6 × 4 | 34.4 / 12 | +2.0 (−4.7, 8.8) | 35.9 / 25 | −0.8 (−6.5, 4.9) | 15 | +19.0 s |
| PhaseNet, centered | e.u., 0.4 × 8 | 42.8 / 23 | −6.4 (−15.4, 2.3) | 41.2 / 35 | −6.1 (−13.5, 1.0) | 31 | |
| EQTransformer, 0 s | e.u., 0.8 × 2 | 11.0 / 13 | +25.5 (15.3, 35.4) | 22.7 / 15 | +12.4 (2.9, 21.5) | 7 | |
| EQTransformer, 1 s | e.u., 0.8 × 2 | 11.1 / 11 | +25.3 (14.9, 35.6) | 20.4 / 17 | +14.7 (6.0, 23.7) | 8 | +17.5 s |
| EQTransformer, 1 s | e.d., 0.4 × 8 | 28.9 / 37 | +7.5 (−4.3, 19.8) | 26.2 / 43 | +8.9 (1.2, 16.8) | 19 | +20.5 s |
| EQTransformer, 2 s | e.u., 0.7 × 8 | 18.9 / 6 | +17.5 (6.7, 28.1) | 20.1 / 12 | +15.0 (5.6, 24.1) | 8 | +21.5 s |
| EQTransformer, 2 s | e.d., 0.5 × 4 | 28.2 / 32 | +8.2 (−2.5, 19.0) | 28.3 / 44 | +6.8 (−1.0, 14.7) | 18 | +19.5 s |
| EQTransformer, centered | both, 0.5 × 8 | 32.2 / 11 | +4.2 (−6.7, 15.3) | 33.4 / 25 | +1.7 (−6.2, 9.7) | 11 | |

Rules: e.u., equal unmatched alarms; e.d., equal detections; setting, threshold × windows. Corr./unm.: chance-corrected detections and unmatched alarms; difference: the learned detector's chance-corrected detections minus the model's, with the 95% paired bootstrap interval by events. Quiet: alarms on the 102.8 h of AFAD quiet windows. On the tuning replays the 1 s PhaseNet setting selected by the equal-unmatched rule had 52.2 chance-corrected detections with 21 unmatched alarms, against 48.7 with 26 for the learned detector; on the $M_w$ 6.1 data it fell to 35.4. Every quiet-window rate test against the learned detector gives $p \geq 0.15$.
Reproduction: `tools/seisbench_scores.py --mode causal --ahead S` (`--ahead 0` reproduces the causal scores byte for byte), then the grid, selection and evaluation of Section S8.

## S16 Selection sensitivity, match criteria, public event sets and location

**The equal-unmatched rule.** The rule was applied to the 25 unmatched alarms that the learned detector had on the tuning data when it was fixed. The anchoring of P times, introduced later, moved the learned detector to 48.7 chance-corrected detections and 26 unmatched alarms without changing either selection; applied to 26, the rule would select another STA/LTA setting (1–10 Hz, STA 0.5 s, LTA 10 s, ratio 8), with 44.8 chance-corrected detections and 26 unmatched alarms on the tuning data, 26.7 with 3 on the $M_w$ 6.1 data and 14.7 alarms per day on the quiet windows (the open diamond of Figure S2). The learned detector remains ahead on the $M_w$ 6.1 data by about ten events rather than 15, against a STA/LTA with fewer unmatched alarms.

**Anchoring.** Anchoring changes only the time attached to a detection, but association and catalog matching use that time, and it can therefore change which detections form an alarm and which catalog event an alarm is matched to. On the tuning replays it moved the adopted rule from 50.0 to 48.7 chance-corrected detections and from 25 to 26 unmatched alarms. Moving the picker window to the anchored P time as well increased the median epicenter error on the Marmara replay from 9.4 to 10.3 km, and the picker therefore keeps the detector's clock.

**Match criteria and close events.** Replayed without the picker, and so without locations, the learned detector keeps every catalog match on all four event replays, and STA/LTA loses 2 of its 30 matches on Marmara; matches rest on the P-time criterion. Because later detections within 40 s at a station are absorbed, an event that closely follows another is harder to detect. Of the 188 catalog events, 24 of the event replays follow the previous one within 40 s, and of these the learned detector detects 9 and the two STA/LTA settings 7 and 14.

**The public KOERI event sets.** On the Sındırgı $M_w$ 4.9 window, recorded by six KO stations, the $M_w$ 4.9 alarms at +18.0 s, as on the AFAD stations, and 8 of 9 catalog events are detected with 2 unmatched alarms. On the Marmara $M_w$ 6.2 and the following hour, recorded by eight KO stations, the $M_w$ 6.2 alarms at +10.5 s and the $M_w$ 5.9 at +11.0 s, and 28 of 37 catalog events are detected. Because 11 of the 15 KO stations contributed training or validation windows to the detector, these sets reproduce the pipeline's behavior on public data but cannot test its generalization; marmara_ko moreover overlaps the detector's training windows (134 windows at CTKS, SLVT, BOTS and BGKT) and is a demonstration only.

**Location.** Location serves the report that follows an alarm and is not part of the evaluation. It is good where several stations surround the source and poor on the $M_w$ 6.1 replay, where only two of six stations lie within 100 km. There, half of the located events are within 10 km and the rest tens of kilometers off (Table S16).

**Table S16:** Epicenter error of the located, catalog-matched events, against the AFAD catalog.

| Replay | Matched | Located | Median | Interquartile range | Within 10 km |
|---|---|---|---|---|---|
| Sındırgı 30 min | 7 | 7 | 5.3 km | 4.7–6.1 km | 5 |
| Marmara | 44 | 35 | 10.6 km | 6.5–24.6 km | 17 |
| Sındırgı $M_w$ 6.1 | 39 | 27 | 32.2 km | 2.8–67.9 km | 13 |
| Sındırgı $M_w$ 4.9, KO | 8 | 7 | 3.7 km | 3.5–3.7 km | 6 |
| Marmara $M_w$ 6.2, KO | 28 | 20 | 6.2 km | 3.8–33.5 km | 12 |

Reproduction: `tools/scorecard.py` on each replay; the selection sensitivity from `tools/select_setting.py` with `--unmatched 26`.

## S17 The chance level of each detector

The chance level $c$ of main-text Equation 2 is the share of catalog events matched when every catalog time is shifted by ±127, ±263 or ±391 s. Inside an aftershock sequence most shifted events land within 60 s of a real one (52 of 69 per shift on the $M_w$ 6.1 replay, 48 of 62 on Marmara), where a true detection of the real event can match them; $c$ then measures chance matching where alarms are densest. Table S17 gives $c$ for every real-time detector on every event replay, as used and with the shifted events that land within 60 s of a real one removed. Without them $c$ falls for every detector; it falls most for the detectors with the most true detections, and the chance-corrected counts rise by 1.5 to 3.5 events. The paired differences between detectors move by at most 0.8 events ($M_w$ 6.1 against STA/LTA at equal unmatched alarms, 14.7, then 14.6 with 30 s and 14.6 with 60 s removed), and the correction is therefore conservative for every detector and favors none.

**Table S17:** Chance level $c$ per detector and replay: as used / without the shifted catalog events that land within 60 s of a real one.

| Detector | Sındırgı 30 min | Marmara | Sındırgı $M_w$ 6.1 | Simav |
|---|---|---|---|---|
| learned | 0.017 / 0.000 | 0.151 / 0.092 | 0.087 / 0.008 | 0.092 / 0.021 |
| STA/LTA, equal unm. | 0.033 / 0.000 | 0.142 / 0.106 | 0.075 / 0.019 | 0.071 / 0.015 |
| STA/LTA, equal det. | 0.037 / 0.056 | 0.275 / 0.209 | 0.145 / 0.071 | 0.139 / 0.060 |
| FilterPicker, equal unm. | 0.017 / 0.000 | 0.174 / 0.082 | 0.058 / 0.000 | 0.101 / 0.021 |
| FilterPicker, equal det. | 0.017 / 0.000 | 0.227 / 0.175 | 0.100 / 0.008 | 0.178 / 0.079 |
| PhaseNet, causal | 0.021 / 0.000 | 0.137 / 0.068 | 0.083 / 0.000 | 0.113 / 0.052 |
| EQTransformer, equal unm. | 0.021 / 0.056 | 0.128 / 0.095 | 0.055 / 0.023 | 0.068 / 0.016 |
| EQTransformer, equal det. | 0.201 / 0.178 | 0.270 / 0.229 | 0.218 / 0.119 | 0.158 / 0.097 |
| shifted events | 59 / 29 | 344 / 86 | 400 / 106 | 338 / 137 |

*Note.* $c$ is the mean over the six shifts of the share of shifted events detected; the last row counts the shifted catalog events over all six shifts. On the 30-minute replay few events remain, and $c$ moves in steps of one event.

Reproduction: `tools/chance_sensitivity.py report` and `filter` on the per-event outcome files, then `tools/significance.py`.

## S18 The trigger grid and the STA/LTA comparison in detail

Table S18a gives a section of the learned detector's trigger grid on the three event replays of the original protocol, and Table S18b the learned detector and the two selected STA/LTA settings on the tuning and $M_w$ 6.1 data, with the alarm times of the largest events and the detections per magnitude band. The STA/LTA settings were selected over 648 combinations of pass band (1–10, 2–10 or 2–20 Hz), STA length (0.5 or 1 s), LTA length (10, 30 or 60 s), trigger ratio (3, 4, 5, 6, 8 or 10), re-arm ratio (1, 1.5 or 2) and characteristic function (vertical or three-component). The re-arm ratio changes the counts by at most two detections and seven unmatched alarms anywhere in the grid and was fixed at 1.5. The chance correction was computed for the 20 candidates selected on uncorrected counts, which bound the corrected counts from above, and every setting that could have been selected after correction was therefore among the candidates.

**Table S18:** Trigger grid and the STA/LTA comparison.

**(a)** Detected/catalog events, chance-corrected detection rate and alarms without a catalog event, for the persistence threshold and window count; the instant branch is at $p \geq 0.9$ throughout, and the first row is the single-window rule.

| Threshold | Windows | Sındırgı 30 min | Marmara | Held out |
|---|---|---|---|---|
| 0.9 | 1 (instant only) | 5/9, 56%, 1 | 39/57, 64%, 10 | 24/66, 33%, 2 |
| 0.85 | 4 | 6/9, 64%, 6 | 46/57, 75%, 52 | 51/66, 73%, 20 |
| 0.85 | 8 | 6/9, 67%, 2 | 44/57, 74%, 16 | 32/66, 45%, 2 |
| 0.8 | 4 | 7/9, 73%, 9 | 47/57, 74%, 80 | 52/66, 73%, 44 |
| **0.8** | **8 (adopted)** | **7/9, 78%, 2** | **44/57, 73%, 24** | **39/66, 55%, 7** |
| 0.8 | 12 | 5/9, 56%, 1 | 42/57, 70%, 10 | 25/66, 35%, 2 |
| 0.7 | 12 | 6/9, 67%, 1 | 42/57, 69%, 15 | 25/66, 35%, 2 |

**(b)** Tuning replays (Sındırgı 30 min and Marmara, 66 catalog events) and the Sındırgı $M_w$ 6.1 replay (66 events). Det.: detected; Corr.: chance-corrected; Unm.: unmatched alarms; alarm times after origin. Of the six M4–5 $M_w$ 6.1 events the learned detector detects five and both STA/LTA settings six.

| | Tuning | | | | Sındırgı $M_w$ 6.1 | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| Detector | Det. | Corr. | Unm. | $M_w$ 6.2 alarm | Det. | Corr. | Unm. | M2–3 | M3–4 | $M_w$ 6.1 alarm |
| learned ($0.8 \times 8$, or 0.9) | 51 | 48.7 | 26 | +11.0 s | 39 | 36.4 | 7 | 12/27 | 21/32 | +17.0 s |
| STA/LTA, equal unmatched | 38 | 33.5 | 25 | +10.0 s | 25 | 21.7 | 9 | 4/27 | 14/32 | +15.7 s |
| STA/LTA, equal detections | 59 | 56.3 | 63 | +9.9 s | 42 | 37.9 | 26 | 15/27 | 20/32 | +15.7 s |

Reproduction: `tools/sweep_trigger.py` on the cached window probabilities and on the STA/LTA replays; `tools/select_setting.py` applies the two rules.

## S19 Training data, phase picker and offline detectors

**Training data.** The detection dataset consists of 6 s three-component windows from 181 stations of the KO, 6G and IJ networks (156, 17 and 8 stations), retrieved from the KOERI FDSN service for every station within 0.5° of cataloged events of $M \geq 2$. Earthquake windows start 2 s before the IASP91 P arrival from the catalog hypocenter, and noise windows are cut from 300 s records beginning 3 h before each event, with the two classes balanced 1:1 and the noise windows selected so that their amplitude distribution, measured as the raw-count standard deviation of the loudest component, matches that of the earthquake windows. The dataset was partitioned by station, and all windows of a station, of either class, belong to one of the training, validation and test sets. The one exception is two earthquake windows of the training stations KRBG and MDNY, whose network code was missing in the source records and which were assigned to the validation and test sets, a change of at most $10^{-4}$ in the test ROC-AUC. Because about 97% of the stations contribute windows of both classes, allocating the classes independently would have placed nearly every station in training under one label and in evaluation under the other. No station of the dataset belongs to the TU archive. Of the 15 stations of the public KOERI replays, 9 belong to the training set, 2 to the validation set and 4 to the test set (Table S1a). The detector was trained with a dropout rate of 0.4 and early stopping, and the three seeds' probabilities are averaged.

**Phase picker.** Location relies on the waveform arm of a P/S picker whose dual-channel design follows Wang and Zhao (2025); in an ablation its second, image-based arm added nothing to pick accuracy. The waveform arm comprises three strided convolutions (kernels of 9, 9 and 7 samples, each followed by batch normalization and ReLU), average pooling to 250 segments of 0.24 s, a bidirectional LSTM, four-head self-attention and a per-segment two-layer classifier over the classes noise, P and S (94 659 parameters). It was trained with soft, PhaseNet-style targets on 30 000 earthquake traces from chunk 2 of STEAD for four epochs with a single seed, using a station-disjoint split of 137, 33 and 32 receivers; on the held-out receivers the median absolute deviation of its S picks is 0.07 s. In the pipeline each 60 s window is detrended, tapered, band-passed and standardized per component, without the asinh transform, and the P and S picks are the segments of highest P and S probability. Because the training data contain no noise traces, the picker returns picks with probabilities of about 0.7 even for pure noise. It is therefore run only after the detector has triggered, and a pick is accepted only if both probabilities are at least 0.5, S follows P by 0–60 s, and P lies between 3 s before and 9 s after the start of the triggering window. Location is a two-stage grid search ($\pm 3°$ at 0.05°, then $\pm 0.1°$ at 0.005°) at 10 km depth in a half-space ($V_P = 6.0$, $V_S = 3.5$ km/s), the worst station being removed while the RMS residual exceeds 2 s.

**Offline detectors and station scaling.** The offline networks of the continuous-record ranking normalize each channel by the long-term mean and standard deviation of its station, accumulated over six archive chunks spread evenly across the record and cleaned in 1 h pieces with the detrending, taper and band-pass applied to the windows. The two P-wave detectors were trained on a separate station-disjoint KOERI dataset of 111 190 windows (120, 28 and 35 stations) that extend from 2.0 s before to 1.4 s after the predicted P arrival, and differ only in the selection of their noise windows. The offline 6 s detector has the corpus and split of the pipeline's detector with station scaling in place of per-window standardization (curated test ROC-AUC 0.9896). The pipeline's per-window detector, scored on the same disjoint windows, reproduces the pipeline's scores to a median of $2 \times 10^{-6}$ (maximum $1.3 \times 10^{-4}$ over 3 702 windows).

## S20 Development results in full

Before the release, the two AFAD sequences of main-text Table 1 (Sındırgı $M_w$ 6.1 and Simav $M_w$ 4.8) and the 23 quiet windows served as this study's test bed, and the settings in use then were the ones selected on the two tuning windows alone. Those results fixed the method, from the trigger rule to the third selection rule of the retuning, and the test protocol counts them as development data. They are given here as they stood, at the pre-retuning settings; the retuned settings of main-text Table 3 replaced them before the tests.

**Trigger rule.** Relative to the single window at $p \geq 0.9$, the adopted rule detects 39 instead of 24 of the 66 $M_w$ 6.1 events, at 7 instead of 2 unmatched alarms (Section S18). The gain lies in M2–4 events, and large events, which saturate the detector and fire the instant branch, keep their alarm times ($M_w$ 6.2 at +11.0 s, $M_w$ 6.1 at +17.0 s and $M_w$ 4.9 at +18.0 s after origin). On the $M_w$ 6.1 replay 8, 10 and 12 windows detect 39, 29 and 25 events, so that the adopted setting lies at the edge of a steep decline. Over the full grids (Figure S2a), the learned detector's envelope lies above that of STA/LTA at every count of unmatched alarms from 1 upward except 14, where both reach 38.6. A single threshold just below the plateau ($p \geq 0.895$) performs comparably but is as sensitive to the plateau's height (Section S10).

**Learned detector versus STA/LTA.** STA/LTA settings were tuned over 648 combinations, and two rules fixed before the run selected one setting at equal unmatched alarms (2–20 Hz, 1 and 30 s, ratio 8, vertical) and one at equal detections (2–10 Hz, 0.5 and 10 s, ratio 6, vertical). At comparable unmatched alarms (7 and 9), the learned detector detects 68% more $M_w$ 6.1 events, with 36.4 against 21.7 chance-corrected detections, a difference of 14.7 (95% interval 5.9–23.7; 7.9–22.2 with 10-minute blocks; exact McNemar $p = 0.003$). The difference lies in M2–4 events (12 against 4 of 27 M2–3 events and 21 against 14 of 32 M3–4 events), whereas STA/LTA detects all six M4–5 events and the learned detector five. At comparable detections (37.9 against 36.4), STA/LTA raises about three times the unmatched alarms, 26 against 7 on the $M_w$ 6.1 replay and 63 against 26 on the tuning data (Table S20a, Section S18). STA/LTA alarms 1.0–1.3 s earlier on the two largest events, because it triggers on the onset sample whereas the window detector must see the P wave inside a 6 s window.

**Other detectors and the Simav sequence.** For each published model and for FilterPicker, the weights and trigger setting were selected on the tuning replays by the two rules above, and both rules chose the INSTANCE weights for the published models. The Simav sequence was replayed once with every setting fixed, on the six archive stations within 300 km, against 56 cataloged events, 39 of them below M2 (Table S20a, Figure S3). On Simav the direction of the STA/LTA comparison repeats, but not its size: at comparable unmatched alarms (17 and 14) the learned detector detected 35.1 chance-corrected events against 29.1, a difference of 6.0 whose interval includes zero, and the gain again lies in M2–4 events. Most of Simav's events are below M2, the lower limit of the detector's training events. Pooled over both sequences, with the chance correction and bootstrap within each, the difference is 20.7 (8.8–32.8; exact McNemar $p = 0.001$), a summary that neither protocol specified. The gain over STA/LTA is not specific to the pipeline's detector: causal PhaseNet matched it on both sequences (pooled difference 1.4, −8.1 to 11.0), with 12 and 28 unmatched alarms against 7 and 17 and with 15 quiet-window alarms against 16, whereas causal EQTransformer was not competitive at equal unmatched alarms. With look-ahead, PhaseNet detected the most events of any detector, but only offline and at three times the unmatched alarms and twice the quiet-window rate of the learned detector (7.2 against 3.74 per day, exact $p = 0.04$); ending the input 1 or 2 s after the window left PhaseNet level with the learned detector and EQTransformer far below it (Section S15). FilterPicker is the strongest classical detector tested: at equal unmatched alarms it detected fewer $M_w$ 6.1 events than the learned detector (difference 7.6, 1.4 to 14.1; McNemar $p = 0.021$) and as many on Simav, and at

equal detections it matched the learned detector with two to three times its unmatched alarms. On the quiet windows its two settings raised 82.4 and 207.1 alarms per day, 22 and 55 times the learned detector's rate, mostly at two stations, CMH and KIRK.

**False alarms on quiet data.** Over 102.8 h without a cataloged event within 250 km, the learned detector raised 16 alarms, 3.74 per day (95% interval 2.14–6.07), against 323, 75.4 per day (67.4–84.1), for STA/LTA at equal unmatched alarms (rate ratio 0.050, 0.028–0.082). That ratio of about 20 belongs to the two selected settings, because STA/LTA was matched on unmatched rather than false alarms; the third selection rule of the retuning answers it. Over the full grids (Figure S2b), at equal tuning detections, the quietest STA/LTA setting raises 5 to 29 times the alarms of the quietest learned setting (4.9 against 0.93 per day at 30 chance-corrected detections, 14.7 against 1.63 at 44 and 60.5 against 2.1 at 48.7). The rate of STA/LTA is similar in the $M_{\mathrm{w}}$ 6.1 window (72 per day) and on quiet data, whereas the learned detector's falls from 56 to 3.74 per day, which suggests that most of its unmatched alarms in the event windows were uncataloged aftershocks. On 5.8 h of public KOERI quiet data the learned detector raised 1 alarm against 25 for STA/LTA. The 16 alarms were classified by criteria that need no judgment of the waveforms (Table S20b, Section S11). Nine were declared with KIRK, whose detection fell 58.6–58.8 s after a 10-minute mark, the slot of a periodic instrumental pulse, a single 0.3 s pulse locked to UTC (Figure S4), confined to May–June 2024 and absent from the event replays; a faulty station raises the network's alarm rate through chance coincidences with the other stations' triggers. A rule fixed before it was run confirms a seismic source when a published picker finds P and S at two declaring stations and the two S−P distances fit the station separation and the P-time difference; it confirms four alarms of the learned detector, one an M2.4 listed by EMSC but not by AFAD, and three of the 323 STA/LTA alarms, all among the same events. Without them the rates are 2.80 and 74.7 per day, and without KIRK the learned detector's rate would be 1.40 per day. Of the 323 STA/LTA alarms, 263 rest on signals shorter than 3 s at both stations, most of them spikes under 0.5 s.

**Alert latency and station subsets.** For the Marmara $M_{\mathrm{w}}$ 6.2, the alarm came 11.0 s after origin, and the S wave reached station ELBA (38 km) 2.1 s later. Over every subset of stations that declared the $M_{\mathrm{w}}$ 6.2, the alert exceeds the P travel time to the second-nearest station by a median of 2.05 s (interquartile range 0.6–2.8 s), and two stations alone, ARNA and ELBA, give the same alert as all eight (Section S5).
Reproduction: stages 5 to 8 of `tools/reproduce_all.sh`; `tools/sweep_trigger.py`, `tools/select_setting.py` and `tools/quiet_sweep.py` for the grids.

**Table S20:** The development results at the pre-retuning settings.

**(a)** Every detector at the settings selected on the two tuning windows: the two AFAD sequences, the quiet windows and the public KOERI event sets.

| | Sındırgı $M_w$ 6.1 (66) | | | Simav $M_w$ 4.8 (56) | | | Quiet | KO (46) |
|---|---|---|---|---|---|---|---|---|
| Detector, rule | Corr. | Unm. | Difference | Corr. | Unm. | Difference | per day | Det./Unm. |
| learned | 36.4 | 7 | | 35.1 | 17 | | 3.74 | 36/25 |
| STA/LTA, equal unm. | 21.7 | 9 | +14.7 (5.9, 23.7) | 29.1 | 14 | +6.0 (−2.4, 14.3) | 75.4 | 23/5 |
| STA/LTA, equal det. | 37.9 | 26 | −1.5 (−9.7, 6.6) | 40.9 | 41 | −5.8 (−12.5, 0.8) | 127.0 | 39/17 |
| FilterPicker, equal unm. | 28.8 | 8 | +7.6 (1.4, 14.1) | 33.8 | 18 | +1.3 (−7.4, 9.8) | 82.4 | 31/12 |
| FilterPicker, equal det. | 34.9 | 16 | +1.5 (−5.6, 8.7) | 39.0 | 49 | −3.9 (−12.7, 4.7) | 207.1 | 36/32 |
| PhaseNet, both | 33.3 | 12 | +3.1 (−4.0, 10.5) | 36.8 | 28 | −1.8 (−8.2, 4.1) | 3.5 | 32/15 |
| EQTransformer, equal unm. | 11.0 | 13 | +25.5 (15.3, 35.4) | 22.7 | 15 | +12.4 (2.9, 21.5) | 1.6 | 24/16 |
| EQTransformer, equal det. | 34.0 | 50 | +2.4 (−9.1, 13.9) | 32.3 | 55 | +2.8 (−5.5, 11.1) | 6.1 | 34/37 |
| *Look-ahead, offline only* | | | | | | | | |
| PhaseNet, equal unm. | 42.8 | 23 | −6.4 (−15.4, 2.3) | 41.2 | 35 | −6.1 (−13.5, 1.0) | 7.2 | 38/23 |
| PhaseNet, equal det. | 38.8 | 9 | −2.3 (−9.2, 4.3) | 41.4 | 27 | −6.3 (−13.1, 0.3) | 3.3 | 32/14 |
| EQTransformer, both | 32.2 | 11 | +4.2 (−6.7, 15.3) | 33.4 | 25 | +1.7 (−6.2, 9.7) | 2.6 | 35/15 |

Corr.: chance-corrected detections; Unm.: unmatched alarms; difference: learned detector's chance-corrected detections minus other's, with 95% paired bootstrap interval by events. Quiet: alarms per day on 102.8 h of AFAD quiet windows. KO: detected events and unmatched alarms on demo_ko and marmara_ko together. Settings: STA/LTA as above; FilterPicker windows 6/10/0.4 s with thresholds 15 and 10 (equal unm.) and 3/5/0.2 s with 4 and 10 (equal det.); published models with INSTANCE weights at threshold × windows $0.7 \times 2$ (PhaseNet), $0.8 \times 2$ and $0.3 \times 8$ (EQTransformer), and with look-ahead $0.4 \times 8$, $0.6 \times 8$ (PhaseNet) and $0.5 \times 8$ (EQTransformer). These are the pre-retuning settings; those of the tests are in main-text Table 3.

**(b)** Alarms of the learned detector on the AFAD quiet windows (102.8 h), by criteria that need no judgment of the waveforms.

| Class | Alarms | Criterion |
|---|---|---|
| KIRK's instrumental pulse | 9 | KIRK declared; its detection in pulse slot |
| seismic, picker-confirmed | 4 | S−P test with PhaseNet or EQTransformer picks |
| undetermined | 3 | neither |
| all alarms | 16 | 3.74 (2.14–6.07) per day |
| excluding picker-confirmed | 12 | 2.80 (1.45–4.89) per day |
| KIRK's pulse only | 9 | 2.10 (0.96–3.99) per day |
| *Sensitivity: picker rule on any associated pair of stations* | | |
| pulse / seismic / undetermined | 8 / 5 / 3 | alarm 1 confirmed through BAND–CMH |
| excluding picker-confirmed | 11 | 2.57 (1.28–4.60) per day |

Upper rows: classification committed before it was run; last two rows: the same picker rule on every pair of stations associated with an alarm, also specified before it was run. Rates: alarms per day with exact Poisson 95% intervals. Each alarm is listed in Table S11.

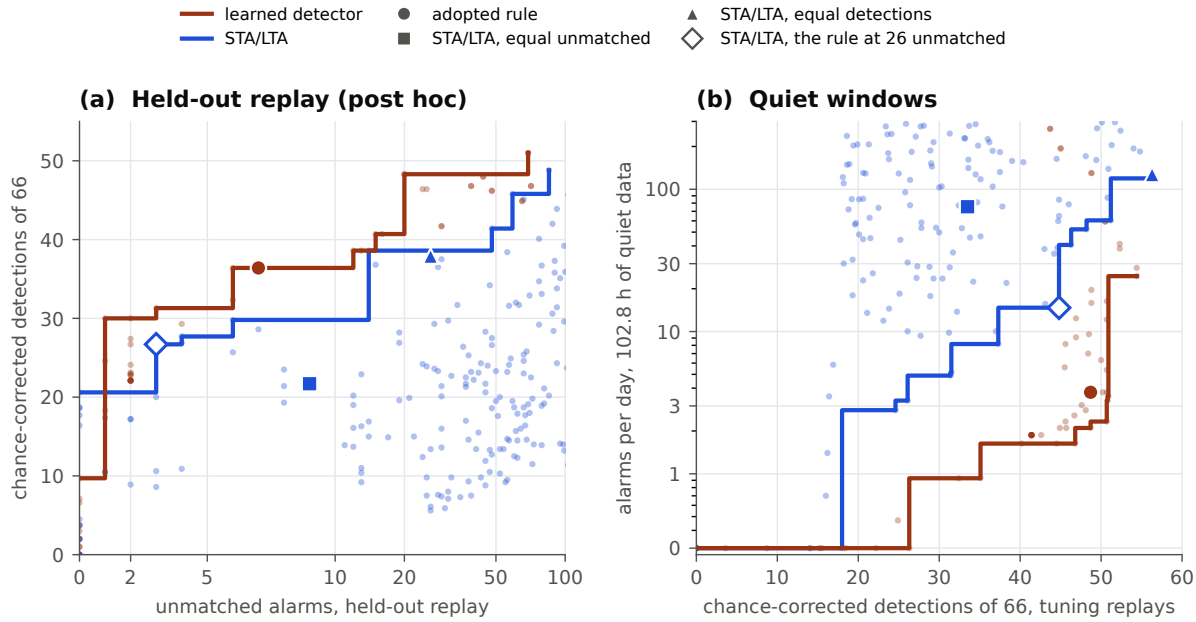



**Figure S2:** Every trigger setting of both detectors (60 of the learned detector, and the 216 of STA/LTA's 648 that have the re-arm ratio of 1.5 at which it was fixed, Section S18; dots), with each detector's envelope (lines) and the pre-retuning selected settings (markers). (a) Sındırgı $M_w$ 6.1 replay, post hoc: chance-corrected detections against unmatched alarms. (b) Alarms per day on 102.8 h of quiet data against chance-corrected detections on the tuning replays; settings without an alarm are drawn on the bottom edge. Open diamond: the STA/LTA setting that the equal-unmatched rule selects when applied to 26 rather than 25 unmatched alarms (Section S16).

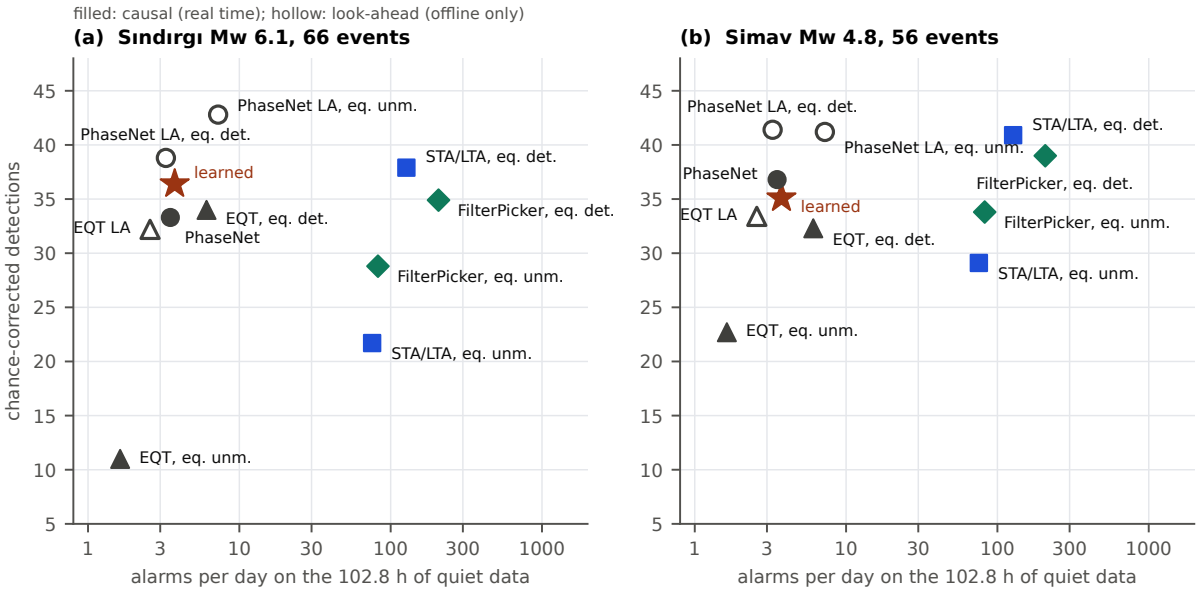



**Figure S3:** Every detector of Table S20a: chance-corrected detections on each AFAD sequence against alarms per day on 102.8 h of quiet data (logarithmic). Up and to the left is better. Filled markers run in real time; hollow markers use input past the window and cannot.

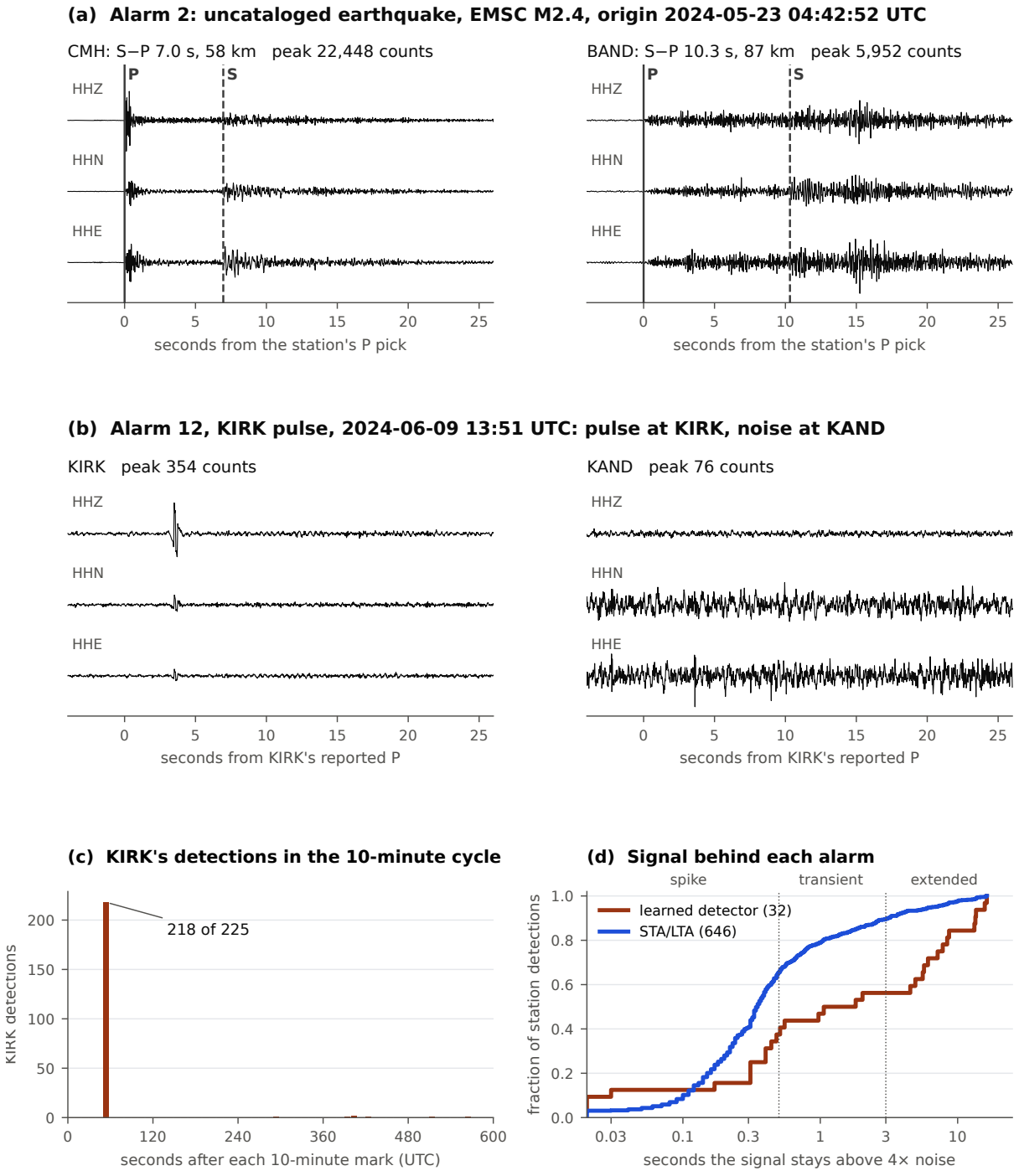



**Figure S4:** Alarms of the learned detector on the AFAD quiet windows. (a) Alarm 2, an uncataloged earthquake (M2.4 in the EMSC catalog; origin time from EMSC) recorded at both declaring stations, with the pipeline's P (solid) and S (dashed) picks. (b) Alarm 12: KIRK's instrumental pulse, about 0.3 s without S wave or coda, coinciding with background noise at KAND. Records band-passed 1–20 Hz. (c) KIRK's 225 detections by time after each 10-minute mark in UTC (10 s bins). (d) Duration of the signal behind each alarm at each declaring station, as the time its envelope stays above four times the noise of the preceding 25 s, for the learned detector's 16 alarms and STA/LTA's 323.

## S21 The selected test sets

`heldout select` of the held-out repository applied the rules of main-text Section 3.1 to each pool after the release and wrote `selection.json`: every qualifying sequence with its coin, stations and scorable count, the stopping point and every quiet pass. Table S21 lists the selected sequences; the quiet windows (24 for Greece and New Zealand, 14 for KO) are in each `selection.json`, released with the pipeline (`docs/heldout-selections/`). Stations are given by code without their network prefix; the two Greek stations that delivered no data in their window, ZKR in sequence 004 and KOKK in sequence 020, are listed as selected and absent from the replay, as each set's `dataset.json` records. KO's pool held three qualifying sequences; the coin assigned the first (21 September 2026, M4.9) to the reserve.

**Table S21:** The selected sequences of the three test networks.

**(a)** Greece: the selected sequences. First event (UTC, catalog magnitude and depth), stations of the replay, scorable events and the running total.

| Seq. | First event (UTC) | $M$ | Depth (km) | St. | Stations | Scorable | Total |
|---|---|---|---|---|---|---|---|
| 002 | 2023-03-21 00:20:14 | 4.6 | 18 | 8 | EVR TETR AMPL PDO PVO LKD2 NYDR TSLK | 41 | 41 |
| 003 | 2023-03-31 16:05:18 | 4.8 | 123 | 8 | EPID KARY MDRA MRTH VILL ATH PTL VLI | 3 | 44 |
| 004 | 2023-04-26 20:00:36 | 5.2 | 108 | 6 | STIA APE NISR NISR2 THERA ZKR | 2 | 46 |
| 008 | 2023-11-03 06:26:37 | 5.1 | 11 | 8 | DOMV MDRA MRTH THIV VILL LKR NEO XOR | 17 | 63 |
| 009 | 2024-01-16 05:20:51 | 4.8 | 14 | 8 | ACOR EPID LOUT THAL VLI GUR LTK VLX | 15 | 78 |
| 010 | 2024-01-19 14:14:13 | 4.8 | 14 | 8 | KARY MDRA MRTH THIV VILL ATH LKR PTL | 21 | 99 |
| 011 | 2024-03-29 07:12:48 | 5.8 | 30 | 8 | ITM ORTH RLS AMT AXS DRO LTHK VLX | 15 | 114 |
| 013 | 2024-07-02 19:34:32 | 4.6 | 6 | 8 | CHAN IPPL KNDR KNSS PFKS STIA IMMV SIVA | 5 | 119 |
| 014 | 2024-07-06 17:34:29 | 4.9 | 11 | 8 | ORTH RLS VLS AMT AXS DRO LTHK PLEV | 24 | 143 |
| 017 | 2024-07-26 16:02:27 | 4.5 | 14 | 8 | CHAN IPPL KNSS PFKS STIA SIVA THERA ZKR | 6 | 149 |
| 020 | 2024-09-22 09:50:23 | 4.6 | 14 | 8 | PLG HORT KAVA KOKK OUR PAIG SOH THAS | 29 | 178 |

**(b)** New Zealand: the selected sequences. First event (UTC, catalog magnitude and depth), stations of the replay, scorable events and the running total.

| Seq. | First event (UTC) | $M$ | Depth (km) | St. | Stations | Scorable | Total |
|---|---|---|---|---|---|---|---|
| 002 | 2023-02-02 13:02:43 | 4.8 | 6 | 7 | HIZ KPRZ KUZ MKAZ RATZ TLZ URZ | 11 | 11 |
| 003 | 2023-02-03 23:14:22 | 5.0 | 86 | 8 | BKZ HIZ KPRZ RATZ TLZ TSZ URZ VRZ | 3 | 14 |
| 004 | 2023-03-17 15:31:06 | 4.7 | 3 | 7 | BKZ KPRZ MWZ PUZ RATZ TLZ URZ | 92 | 106 |
| 008 | 2023-10-15 03:15:20 | 4.6 | 202 | 6 | BKZ HIZ KPRZ RATZ TLZ VRZ | 2 | 108 |
| 009 | 2023-10-23 09:09:04 | 5.1 | 5 | 5 | DSZ KHZ RPZ WHSZ WVZ | 3 | 111 |
| 010 | 2024-02-18 04:09:10 | 4.8 | 175 | 8 | BKZ KPRZ KUZ MKAZ MWZ RATZ TLZ URZ | 5 | 116 |
| 011 | 2024-02-26 08:21:12 | 5.1 | 5 | 5 | CVZ FOZ RPZ WHSZ WVZ | 7 | 123 |
| 013 | 2024-04-02 05:39:44 | 4.5 | 5 | 5 | CVZ FOZ RPZ WHSZ WVZ | 4 | 127 |
| 014 | 2024-04-22 21:57:41 | 4.6 | 39 | 8 | BKZ KPRZ MWZ RATZ TLZ TSZ URZ VRZ | 9 | 136 |
| 017 | 2024-11-26 14:46:49 | 4.9 | 198 | 7 | BKZ HIZ KPRZ RATZ TLZ TSZ VRZ | 4 | 140 |
| 020 | 2025-07-18 11:00:56 | 4.6 | 33 | 6 | BKZ KPRZ MWZ PUZ RATZ URZ | 4 | 144 |
| 023 | 2025-11-01 00:47:37 | 4.8 | 5 | 5 | DSZ KHZ RPZ WHSZ WVZ | 5 | 149 |
| 024 | 2025-11-10 18:51:35 | 4.6 | 211 | 7 | BKZ HIZ KPRZ RATZ TLZ TSZ VRZ | 5 | 154 |

**(c)** KO: the selected sequences. First event (UTC, catalog magnitude and depth), stations of the replay, scorable events and the running total.

| Seq. | First event (UTC) | $M$ | Depth (km) | St. | Stations | Scorable | Total |
|---|---|---|---|---|---|---|---|
| 002 | 2026-09-24 07:40:58 | 5.3 | 7 | 7 | AKDG DYBB GAZ MAZI MLTY NURH SVRC | 7 | 7 |
| 003 | 2026-10-04 11:19:25 | 4.7 | 8 | 6 | BNN CEYT GAZ KARA NURH SARI | 29 | 36 |

Catalog magnitudes are the catalogs' preferred ones (NOA MLh, GeoNet MLv or M, EMSC mw or mb). Scorable: the stopping rule's count of catalog events in the replay within 250 km of the stations' centroid. Reproduction: `heldout select greece|nz|ko --allow-test-period` at the commits of Table S9.

## S22 Quiet-window alarm rates on the test networks

Table S22 gives each detector's alarms on each network's quiet windows (main-text Table 4: 24 windows of 5.5 scored hours on Greece and New Zealand, 14 on KO) and the ratio of its rate to the learned detector's with the exact conditional 95% interval of the two Poisson counts (`tools/quiet_sweep.py`, written by `run_heldout.py` to `analysis/quiet.txt`). With the two stations of most STA/LTA station detections at equal unmatched alarms left out for every detector alike (`tools/quiet_without_stations.py`): on New Zealand, WHSZ and CVZ, the learned detector 3 alarms (0.55 per day) and STA/LTA at equal unmatched alarms 8 (1.45); on KO, GAZ and SVRC, 27 (8.42) and 18 (5.61); on Greece, CHAN and PFKS, 67 (12.18) and 32 (5.82). Events deeper than 40 km with a Pd estimate (main-text Section 4.5): on New Zealand, one MLv 4 to 5.5 event, final error −0.3 for the learned detector, and one smaller event, +1.1.

**Table S22:** Alarms on the quiet windows of each test network: the count, the rate per day and the ratio of the detector's rate to the learned detector's with its exact 95% interval.

| | Greece (132 h) | | | New Zealand (132 h) | | | KO (77 h) | | |
|---|---|---|---|---|---|---|---|---|---|
| Detector, rule | Alarms | Per day | Ratio | Alarms | Per day | Ratio | Alarms | Per day | Ratio |
| learned | 106 | 19.27 | – | 5 | 0.91 | – | 31 | 9.66 | – |
| STA/LTA, 1 | 114 | 20.73 | 0.93 (0.71–1.22) | 9 | 1.64 | 0.56 (0.15–1.85) | 67 | 20.88 | 0.46 (0.29–0.72) |
| STA/LTA, 2 | 571 | 103.82 | 0.19 (0.15–0.23) | 33 | 6.00 | 0.15 (0.05–0.39) | 359 | 111.90 | 0.09 (0.06–0.12) |
| STA/LTA, 3 | 18 | 3.27 | 5.89 (3.55–10.30) | 2 | 0.36 | 2.50 (0.41–26.30) | 11 | 3.43 | 2.82 (1.38–6.21) |
| STA/LTA, per station | 23 | 4.18 | 4.61 (2.92–7.58) | 12 | 2.18 | 0.42 (0.12–1.27) | 8 | 2.49 | 3.88 (1.74–9.76) |
| FilterPicker, 1 | 247 | 44.91 | 0.43 (0.34–0.54) | 34 | 6.18 | 0.15 (0.04–0.38) | 100 | 31.17 | 0.31 (0.20–0.47) |
| FilterPicker, 2 | 1486 | 270.18 | 0.07 (0.06–0.09) | 154 | 28.00 | 0.03 (0.01–0.08) | 647 | 201.66 | 0.05 (0.03–0.07) |
| FilterPicker, 3 | 13 | 2.36 | 8.15 (4.57–15.80) | 5 | 0.91 | 1.00 (0.23–4.35) | 5 | 1.56 | 6.20 (2.39–20.40) |
| PhaseNet causal, 1 | 71 | 12.91 | 1.49 (1.10–2.05) | 2 | 0.36 | 2.50 (0.41–26.30) | 23 | 7.17 | 1.35 (0.76–2.42) |
| PhaseNet causal, 2 | 140 | 25.45 | 0.76 (0.58–0.98) | 8 | 1.45 | 0.62 (0.16–2.17) | 37 | 11.53 | 0.84 (0.50–1.39) |
| PhaseNet causal, 3 | 71 | 12.91 | 1.49 (1.10–2.05) | 2 | 0.36 | 2.50 (0.41–26.30) | 22 | 6.86 | 1.41 (0.79–2.55) |
| EQTransformer causal, 1 | 20 | 3.64 | 5.30 (3.27–9.03) | 0 | 0.00 | inf (0.92–inf) | 5 | 1.56 | 6.20 (2.39–20.40) |
| EQTransformer causal, 2 | 91 | 16.55 | 1.17 (0.87–1.56) | 7 | 1.27 | 0.71 (0.18–2.61) | 37 | 11.53 | 0.84 (0.50–1.39) |
| EQTransformer causal, 3 | 47 | 8.55 | 2.25 (1.59–3.25) | 3 | 0.55 | 1.67 (0.32–10.70) | 14 | 4.36 | 2.21 (1.14–4.50) |
| PhaseNet look-ahead, 1 | 39 | 7.09 | 2.72 (1.87–4.03) | 1 | 0.18 | 5.00 (0.56–236.00) | 15 | 4.68 | 2.07 (1.08–4.12) |
| PhaseNet look-ahead, 2 | 45 | 8.18 | 2.36 (1.65–3.42) | 4 | 0.73 | 1.25 (0.27–6.30) | 22 | 6.86 | 1.41 (0.79–2.55) |
| PhaseNet look-ahead, 3 | 66 | 12.00 | 1.61 (1.17–2.22) | 4 | 0.73 | 1.25 (0.27–6.30) | 24 | 7.48 | 1.29 (0.73–2.30) |
| EQTransformer look-ahead, 1 | 39 | 7.09 | 2.72 (1.87–4.03) | 1 | 0.18 | 5.00 (0.56–236.00) | 11 | 3.43 | 2.82 (1.38–6.21) |
| EQTransformer look-ahead, 2 | 35 | 6.36 | 3.03 (2.05–4.57) | 14 | 2.55 | 0.36 (0.10–1.05) | 20 | 6.23 | 1.55 (0.86–2.87) |
| EQTransformer look-ahead, 3 | 51 | 9.27 | 2.08 (1.47–2.96) | 18 | 3.27 | 0.28 (0.08–0.78) | 29 | 9.04 | 1.07 (0.62–1.84) |

## S23 Detection endpoints for events no deeper than 40 km

Table S23 repeats main-text Table 5 for the events no deeper than 40 km by the catalog (`tools/significance.py --max-depth 40`; the restriction is applied to the per-event outcome tables before any statistic, in the observed and in the shifted runs). New Zealand keeps 137 of 151 events (14 deeper, none without a depth); Greece keeps 175 of 177 (2 deeper); KO keeps all 36. No conclusion changes: on Greece the primary difference is +22.4 (+11.3 to +33.6) and the equivalence with causal PhaseNet holds (+5.0, 90% interval −1.0 to +11.3, margin ±17.5); on New Zealand the primary difference is +17.9 (+7.2 to +29.0) against +18.1 for all events, the equivalence with causal PhaseNet holds (+4.8, 90% interval −2.6 to +12.3, margin ±13.7), the adjacent settings with 6 windows remain ahead of the rule and those with 10 behind it, and the comparison restricted to events detected before S is +15.5 (+4.6 to +26.7). The five New Zealand sequences whose first event is deeper than 40 km (86, 175, 198, 202 and 211 km) contribute most of the 14 events left out.

**Table S23:** Each detector on the test networks, restricted to the events no deeper than 40 km: chance-corrected detections and the difference from the learned detector with its 95% paired bootstrap interval.

| | Greece (175) | | New Zealand (137) | | KO (36) | |
|---|---|---|---|---|---|---|
| Detector, rule | Corr. | Difference | Corr. | Difference | Corr. | Difference |
| learned | 148.3 | – | 80.2 | – | 15.5 | – |
| STA/LTA, 1 | 125.9 | +22.4 (+11.3, +33.6) | 62.2 | +17.9 (+7.2, +29.0) | 13.0 | +2.5 (−0.5, +5.9) |
| STA/LTA, 2 | 136.9 | +11.4 (+2.2, +21.2) | 75.9 | +4.3 (−6.2, +14.3) | 16.6 | −1.1 (−6.9, +4.6) |
| STA/LTA, 3 | 98.3 | +50.0 (+36.9, +63.3) | 37.7 | +42.5 (+31.0, +54.3) | 8.0 | +7.5 (+3.0, +12.7) |
| STA/LTA, per station | 128.4 | +19.9 (+9.3, +30.5) | 51.4 | +28.8 (+17.1, +40.0) | 10.0 | +5.5 (+1.4, +9.9) |
| FilterPicker, 1 | 126.4 | +21.9 (+10.9, +33.1) | 85.2 | −5.0 (−15.3, +5.2) | 11.0 | +4.5 (+0.6, +8.8) |
| FilterPicker, 2 | 131.6 | +16.7 (+5.1, +28.5) | 99.2 | −19.0 (−29.7, −8.4) | 13.7 | +1.8 (−2.6, +6.4) |
| FilterPicker, 3 | 88.7 | +59.6 (+47.0, +72.1) | 55.1 | +25.1 (+14.7, +35.9) | 8.9 | +6.7 (+2.3, +11.7) |
| PhaseNet causal, 1 | 143.3 | +5.0 (−2.1, +12.5) | 75.4 | +4.8 (−3.9, +13.8) | 10.0 | +5.5 (+1.4, +9.9) |
| PhaseNet causal, 2 | 142.5 | +5.8 (−2.4, +14.5) | 95.7 | −15.5 (−25.5, −5.8) | 15.8 | −0.3 (−3.3, +2.7) |
| PhaseNet causal, 3 | 141.9 | +6.4 (−1.5, +14.5) | 84.5 | −4.3 (−13.0, +4.4) | 11.9 | +3.6 (+0.4, +7.7) |
| EQTransformer causal, 1 | 87.9 | +60.4 (+45.7, +75.5) | 23.9 | +56.3 (+44.9, +67.5) | 5.0 | +10.5 (+5.2, +16.1) |
| EQTransformer causal, 2 | 135.3 | +13.0 (+3.0, +23.7) | 62.5 | +17.6 (+7.1, +28.5) | 10.8 | +4.8 (−0.1, +9.9) |
| EQTransformer causal, 3 | 126.0 | +22.2 (+10.5, +34.1) | 41.4 | +38.8 (+27.6, +50.1) | 9.9 | +5.7 (+0.6, +10.9) |
| PhaseNet look-ahead, 1 | 141.9 | +6.3 (−3.0, +16.1) | 83.6 | −3.5 (−13.5, +6.6) | 11.0 | +4.5 (+0.6, +8.8) |
| PhaseNet look-ahead, 2 | 142.1 | +6.2 (−2.9, +15.4) | 86.7 | −6.5 (−16.8, +4.1) | 12.0 | +3.5 (+0.3, +7.6) |
| PhaseNet look-ahead, 3 | 143.6 | +4.6 (−4.9, +14.4) | 92.8 | −12.6 (−23.1, −2.3) | 12.0 | +3.5 (−0.7, +7.9) |
| EQTransformer look-ahead, 1 | 115.5 | +32.8 (+20.7, +45.0) | 29.6 | +50.6 (+37.6, +63.0) | 1.0 | +14.5 (+8.6, +20.6) |
| EQTransformer look-ahead, 2 | 148.6 | −0.4 (−9.6, +8.8) | 88.4 | −8.3 (−18.8, +2.0) | 17.8 | −2.2 (−5.4, +0.0) |
| EQTransformer look-ahead, 3 | 152.7 | −4.4 (−13.7, +4.6) | 95.1 | −14.9 (−25.2, −5.0) | 18.7 | −3.2 (−6.7, −0.2) |

## S24 The test protocol's rules in full, and the secondary endpoints

The protocol's text is `docs/protocol-heldout.md` of release v1.0.0; this section gives the rules the main text summarizes and the secondary endpoints in full.

**Per-station STA/LTA.** The network-wide equal-unmatched setting (main-text Table 3) with each station's trigger ratio replaced by the lowest of 3, 4, 5, 6, 8, 10, 12 and 15 at which the station's trigger rate on four 6 h windows from before the test period (drawn with seed 20221231; Greece and New Zealand in 2022, KO from 1 July to 29 August 2026) does not exceed its network group's target. The rate counts only the time outside the arrivals of cataloged events the station could have recorded; the target is the median station rate of the network-wide setting on the same windows, 0.845, 0.548 and 0.318 triggers per quiet hour for Greece, KO and New Zealand. A station above the target at every ratio gets 15, and a test station without tuning data keeps ratio 8 (`tools/stalta_station_ratios.py`). Before the release the target was changed, on 29 October 2026, from the median station rate on the AFAD quiet windows (0.307 per quiet hour, with the grid ending at 10), against which most Greek and KO stations ended at the grid's top ratio.

**Seeds and counts.** Reserve coin 42 (Python's `random.Random(42)`, one draw per sequence, below 0.5 is this study's); quiet windows 43, prospective quiet windows 44, tuning windows 20221231. The stopping rule counts a sequence's scorable events as the pool catalog's events with origin within the replay and epicenter within 250 km of the stations' centroid, any magnitude, with or without depth; ayzek scores the set's catalog excerpt, which leaves out events without depth or magnitude, with origin from 30 s after the replay's start to 30 s before its end.

**Deviations.** A bug found after the test data are fetched is fixed and the results reported both ways, labeled as a deviation; nothing is retuned on test data. One was found after the three runs, in the analysis tool and not in the pipeline. Each report lists, for the stations of a detected event, the S arrival and the warning before it, the S arrival picked where the pipeline's picker has a pick and predicted otherwise. The tool of the release read only the predicted rows, so the warning-time endpoint (x) left out the stations with a pick, those nearest the source, and on Greece 23 of the learned detector's 153 detected events, every station of which had a pick. The fix (commit 87e692c of the pipeline's repository) reads both kinds of row. Rerun on the same reports, the analysis stage reproduces every other output file byte for byte; Table S24 gives the endpoint both ways. The one tooling fault, an out-of-date build of `tools/network_subsets` that stopped KO's analysis stage before it wrote any output, changed no number (Table S13).

**Prospective tests.** New Zealand and KO, on data recorded from the release date on: every qualifying sequence in time order with no reserve split, until the network's scorable events reach its target (150, 240) or 12 months have passed, whichever comes first, in which case every sequence is used and the result is reported as under target. Once a month, the qualifying sequences whose first event is at least four weeks old are identified from the network's catalog and fetched with the released tool, each once, with its catalog excerpt at that time; later catalog revisions are not taken up. After the span closes, 24 quiet windows are drawn over it with seed 44 on the station sets of its sequences. Detectors, settings, scoring and endpoints are those of the retrospective tests. The archive delivers the post-release data intact, so the test replays them; a live SeedLink run, if made, is reported apart.

**The secondary endpoints in full.** Main-text Section 4.4 condenses the following, written from `analysis/significance.txt`, `quiet.txt`, `quiet_without_stations.txt`, `neighbours.txt`, `significance_pre_s.txt` and `station_split.txt` of each run.

**Table S24:** Warning time before S at the alarm (endpoint x): events with a station still ahead of S out of the detected events with a warning row, and the median warning at the nearest such station, as computed by the release's tool (predicted S only) and after the fix (picked and predicted S).

| | | Release's tool | | Fixed | |
|---|---|---|---|---|---|
| Network | Detector | Events ahead | Median (s) | Events ahead | Median (s) |
| Greece | learned | 126 / 130 | 4.8 | 151 / 153 | 4.5 |
| | STA/LTA, rule 1 | 103 / 104 | 5.1 | 132 / 132 | 5.3 |
| New Zealand | learned | 83 / 83 | 13.0 | 87 / 87 | 7.7 |
| | STA/LTA, rule 1 | 64 / 64 | 14.7 | 69 / 69 | 8.5 |
| KO | learned | 16 / 16 | 5.4 | 16 / 16 | 3.7 |
| | STA/LTA, rule 1 | 13 / 13 | 7.7 | 13 / 13 | 7.6 |

*Note.* KO's $M_w$ 5.3: 16.0 and 18.6 s at AKDG (131 km, predicted S) before the fix, 10.2 and 12.9 s at MLTY (107 km, picked S) after it. Alarm delays after origin do not use these rows and are unchanged. Reproduction: `tools/run_heldout.py --stages analysis` at 87e692c on the release-time reports.

**Against the other baselines (ii, iii).** On New Zealand, STA/LTA at equal quiet alarms detected 40.7 chance-corrected events (difference +43.5, +32.0 to +55.1) and the per-station STA/LTA 54.4 (+29.8, +17.8 to +41.6). FilterPicker at equal unmatched alarms detected as many events as the learned detector (89.2; −5.0, −16.0 to +5.6) but with 67 unmatched alarms against 20, so its development-data setting did not hold its operating point on this network; at equal quiet alarms it detected 58.1 (+26.1). Causal EQTransformer detected 25.9 at equal unmatched alarms (+58.3) and 65.5 at equal detections. The look-ahead settings, which cannot run in real time, reached 85.6 to 104.0 at 20 to 73 unmatched alarms. On KO, with its 36 events, the learned detector was ahead of STA/LTA at equal quiet alarms (+7.5, +3.0 to +12.7), the per-station STA/LTA (+5.5, +1.4 to +9.9), FilterPicker at equal unmatched alarms (+4.5, +0.6 to +8.8) and causal EQTransformer (+10.5, +5.2 to +16.1), and level with the equal-detection settings.

**Equivalence with causal PhaseNet (iv).** On New Zealand the learned detector and causal PhaseNet at equal unmatched alarms differ by +5.8 events (90% interval −1.8 to +13.9) against a margin of ±15.1, so the two are equivalent under the two one-sided tests, with PhaseNet at 15 unmatched alarms against 20. On KO the difference is +5.5 (+1.9 to +9.5) against a margin of ±3.6: not shown equivalent, in the learned detector's favor. On Greece the difference is +5.0 (−0.9 to +11.6) against ±17.7, equivalent, with PhaseNet at 229 unmatched alarms against the learned detector's 371.

**Quiet windows (v).** On New Zealand's 132 scored hours the learned detector raised 5 alarms, 0.9 per day, against 1.6 for STA/LTA at equal unmatched alarms (rate ratio 0.56, exact 95% interval 0.15 to 1.85), 6.0 at equal detections, 0.4 at equal quiet alarms and 2.2 for the per-station STA/LTA; FilterPicker's three settings raised 6.2, 28.0 and 0.9 per day, causal PhaseNet's 0.4 to 1.4 and causal EQTransformer's 0.0 to 1.3. Without WHSZ and CVZ, the two stations with the most STA/LTA detections, the learned detector's rate is 0.5 per day and STA/LTA's 1.5. On KO's 77 scored hours the learned detector raised 31 alarms, 9.7 per day, against 20.9 for STA/LTA at equal unmatched alarms (ratio 0.46, 0.29 to 0.72) and 111.9 at equal detections; but STA/LTA at equal quiet alarms (3.4), the per-station STA/LTA (2.5), FilterPicker at equal quiet alarms (1.6) and causal EQTransformer at equal unmatched alarms (1.6) were all quieter than the learned detector, and causal PhaseNet's rates (6.9 to 11.5) were not distinguishable from it. The learned detector's KO rate is 2.6 times its rate on the AFAD quiet windows; leaving out GAZ and SVRC lowers it only to 8.4 per day while it lowers STA/LTA's to 5.6. On Greece's 132 scored hours the learned detector raised 106 alarms, 19.3 per day, against 20.7 for STA/LTA at

equal unmatched alarms (ratio 0.93, 0.71 to 1.22), 103.8 at equal detections, 3.3 at equal quiet alarms and 4.2 for the per-station STA/LTA; FilterPicker's settings raised 44.9, 270.2 and 2.4, causal PhaseNet's 12.9 to 25.4 and causal EQTransformer's 3.6 to 16.6. Without CHAN and PFKS the learned detector's rate is 12.2 per day and STA/LTA's 5.8; one window holds half of the learned detector's alarms (Section S25).

**Adjacent settings (vi).** On New Zealand the rule's neighbors with 6 consecutive windows instead of 8 detected 11 to 20 more chance-corrected events than the rule (thresholds 0.85, 0.8 and 0.75), and those with 10 windows 8 to 9 fewer; on KO the neighbors lie within 2 events of the rule. On Greece no neighbor detected more than 0.2 events beyond the rule, and those with 10 windows 8.5 to 14.4 fewer. The rule was not retuned.

**Detection before S (vii).** On Greece 322 of the learned detector's 332 alarm detections (97%) preceded the predicted S, 143 of its 153 detected events were detected before S at every station, and the primary difference on them is +15.7 (+4.1 to +27.0). On New Zealand 170 of the learned detector's 176 station detections that formed an alarm (97%) were declared before the predicted S, and 82 of its 87 detected events were detected before S at every such station; restricted to those events, the primary difference is +15.7 (+4.7 to +27.1). On KO every detection of both detectors preceded S.

**KO's station split (viii).** Every station of KO's two sequences is a training-split station of the learned detector, so the comparison on test-split stations is empty; the training-split result is the KO result above.

## S25 Exploratory: the Greek unmatched and quiet-window alarms

This analysis is outside the protocol. It was made after the Greek result, when the learned detector's 371 unmatched alarms (against 189 for STA/LTA at the setting matched on the development data) and its 19.3 quiet-window alarms per day asked for an explanation. Nothing in it changes a number of the protocol's endpoints.

**Unmatched alarms in the two noisiest sequences.** An alarm counts as located if the pipeline gave it a P-time location from at least five stations with a 68% radius of at most 20 km, or a pick location from at least three stations with an rms of at most 1 s, and as located in the sequence if that location lies within 30 km of the sequence's first event. At Epirus (MLh 4.6, 21 March 2023; station AMPL 5 km from the epicenter, the other seven 41 to 52 km), 52 of the learned detector's 121 unmatched alarms locate in the sequence, 43 of them from P and S picks, 5 elsewhere and 64 not at all; AMPL takes part in 77. An event located from eight stations at the epicenter 8 min before the mainshock is absent from NOA's catalog. At Evia (MLh 5.1, 3 November 2023; all stations 51 to 77 km away), 25 of 104 locate in the sequence, 23 from picks, 2 elsewhere and 77 not at all, and the unmatched alarms per half hour rise and decay as an aftershock sequence does (15, 21, 28, 17, 15, 8 from 10 min before the mainshock). The ISC bulletin lists 9 events in the Evia window and 1 in the Epirus window from the Aristotle University network that NOA's catalog lacks, and an unmatched alarm of the learned detector follows each within 60 s. Four of the alarms that follow them are among those located in the sequences, and 7 are not (6 unlocated, 1 located elsewhere). The 225 unmatched alarms therefore divide into 77 located in the sequences, 7 others following an event missing from NOA's catalog, 135 other unlocated alarms and 6 others located elsewhere; at least 84 (the first two groups) are earthquakes missing from the catalog. The unlocated alarms, mostly declared by two stations 40 to 80 km from the source, cannot be classified from the reports.

**The same classification for the other detectors.** If the learned detector's excess of unmatched alarms were mostly sensitivity to small earthquakes, an equally sensitive detector would share it. Causal PhaseNet at rule 1 had the same development operating point (52 unmatched alarms) and detected as many of the Greek cataloged events (Table 5 of the main text). Table S25 classifies the unmatched alarms of the three detectors in the two sequences by the same rule. Of the learned detector's 110 unmatched alarms beyond causal PhaseNet's, 24 locate in the sequences and 84 cannot be located; at Evia station VILL takes part in 50 of the learned detector's unmatched alarms and in 5 of PhaseNet's, and in 38 of its 77 unlocated ones against 3 of PhaseNet's 20. Of the 9 Evia events missing from NOA's catalog, the learned detector alarmed after 9, STA/LTA after 6 and causal PhaseNet after 5. The learned detector thus finds a few more of the small uncataloged events than the other two, but most of its excess over PhaseNet consists of alarms that cannot be located and that PhaseNet, at equal detections of cataloged events, does not raise.

**The quiet windows.** Window 14 (central and eastern Crete, 19 November 2025, 03:20 to 09:20 UTC; stations CHAN, IPPL, KNDR, KNSS, PFKS, STIA, IMMV and SIVA) holds 51 of the learned detector's 106 Greek quiet alarms. STIA takes part in 49, KNSS in 35 and CHAN in 18; STIA triggered 162 times and KNSS 102 in the 6 h, and only 16 of the window's 312 triggers had an STA/LTA onset in their window, so the signals were emergent rather than impulsive. None of the 51 alarms has a pick location, and neither EMSC nor the ISC lists an event within 1.5° in the window. STA/LTA raised 2 alarms there. Without window 14 the learned detector raised 55 alarms in 126.5 h, 10.4 per day, against 112, 21.2 per day, for STA/LTA at equal unmatched alarms. The protocol leaves out the two stations with the most STA/LTA detections, CHAN and

**Table S25:** Unmatched alarms of three detectors (rule 1) in the Epirus and Evia sequences, by where they locate.

| Detector | Unmatched | In the sequence | Unlocated | Elsewhere | After the 9 Evia events missing from NOA's catalog |
|---|---|---|---|---|---|
| Learned | 225 (121 + 104) | 77 (52 + 25) | 141 (64 + 77) | 7 | 9 |
| Causal PhaseNet | 115 (82 + 33) | 53 (42 + 11) | 57 (37 + 20) | 5 | 5 |
| STA/LTA | 114 (54 + 60) | 22 (14 + 8) | 90 (40 + 50) | 2 | 6 |

*Note.* In parentheses, Epirus + Evia. Located in the sequence: within 30 km of NOA's epicenter of the sequence's first event (38.963°N, 21.208°E at Epirus; 38.738°N, 23.581°E at Evia), by the rule above. Last column: an unmatched alarm within 60 s after the event.

PFKS, which does not reach STIA because STA/LTA does not fire on its signals.
Reproduction: `python3 tools/classify_unmatched.py REPORT LAT LON` of the pipeline's repository on the Greek test's reports; the ISC and EMSC listings come from their FDSN event services for each replay's span within 2.25° of the first event (1.5° of 35.2°N, 25.0°E for the Crete window).

## S26 Exploratory: clustering within sequences

This analysis is outside the protocol. The protocol's bootstrap resamples events within each sequence, and the exact McNemar test treats events as independent, but the events of an aftershock sequence share stations, noise and the 40 s coda absorption, and a few sequences hold most of the events: New Zealand's sequence 004 holds 89 of its 151 scored events, and Greece's 002, 020 and 014 hold half of its 177. Table S26 gives the primary comparison with blocks of 10 and 15 min of origin time resampled whole, as on the development data (Section S20), and with each sequence left out in turn.

**Table S26:** The primary comparison (learned detector against STA/LTA at rule 1) under clustering: the difference in chance-corrected detections and its 95% bootstrap interval.

| | | | Blocks resampled | | One sequence left out | |
|---|---|---|---|---|---|---|
| Network | Events | Events resampled | 10 min | 15 min | Difference | Lowest bound |
| Greece | 177 | +22.4 (+11.7, +33.5) | (+11.0, +34.3) | (+10.1, +35.6) | +15.7 to +23.5 | +6.6 |
| New Zealand | 151 | +18.1 (+7.2, +29.5) | (+7.6, +28.6) | (+6.7, +30.0) | +7.4 to +18.1 | +3.2 |
| KO | 36 | +2.5 (−0.5, +5.9) | (−0.4, +5.8) | (−0.4, +5.8) | 0.0 to +2.5 | −0.5 |

*Note.* Greece's lowest difference is without sequence 002 (137 events left), its lowest bound without 014. New Zealand's lowest difference and bound are without sequence 004: +7.4 (+3.2, +12.3) of 62 events, 12% of them as with all 151; without any other sequence its difference is +16.0 to +18.1, the lowest bound +5.1. KO without its sequence 002 keeps 7 events.

Reproduction: `python3 tools/significance.py` on the test's outcome tables with `--against stalta_eqfa`, adding `--block-minutes 10` or `15`, or with one sequence removed from `--groups`.

## References

The three works cited in this supplement, all also in the references of the main text.


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